Bundled Risks, Dollar Index Options, and Quantitative Implications for Dynamic Currency Models*

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October 12, 2023

Abstract

We propose a model of options on futures on the dollar index, which are traded vehicles to protect against concerted U.S. dollar appreciations or depreciations. Estimating the model, we draw quantitative assessments and examine consistency with our empirical findings. First, average excess returns for out-of-the-money (OTM) call options on dollar index futures are negative and statistically significant, whereas those for OTM put options are insignificant. Second, average excess returns for call options become more negative at higher strikes. The estimated model adheres to the data on dollar index option prices, volatilities, and supports negative risk premiums for OTM call options.

JEL: G11, G12, G13, C5, D24, D34.
Key words: Traded bundled risks, model of options on futures on the dollar index, formulation of dollar index option risk premiums, returns of options on futures on the dollar index, dynamic models of currency behavior

*First Draft: April 2022. Posted on SSRN: October 2023. Seminar participants at Temple University provided many suggestions. We thank John Crosby, Jorge Hansen, Steve Heston, Juan Londono, Dilip Madan, Oleg Rytchkov, Sam Rosen, Jinming Xue, and Zhaowei Zhang for many helpful conversations. Full replication codes and data used are available from the authors.

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1. Introduction

In this paper, we examine options on futures on the dollar index, which are vehicles to protect against concerted U.S. dollar appreciations or depreciations. Traded instruments that bundle disperse currency risks are relevant to understanding phenomena through modeling. We contribute by developing a model of futures on the geometrically weighted dollar index and investigating the quantitative implications of options written on the futures on the dollar index. The dollar index is quoted in European terms — the U.S. dollar is the reference — but is settled in U.S. dollars, posing a modeling difficulty (details later).

Some quantitative questions can be more suitably considered by exploiting options on futures on the dollar index. For example, what could be the drivers of the option risk premiums that may underlie collective dollar concerns to the upside of dollar index? If the U.S. dollar provides a cover to the rest of the world by appreciating during periods of bad economic shocks, then holding out-of-the-money call options on the dollar index is valuable. This feature underpins a negative call option risk premium. Even though the currency literature is vast, the data ingredients of futures on the dollar index and option prices on futures on the dollar index are seldom utilized to study risk premium channels or dissecting modeling consistency.

We provide evidence on collective concerns about dollar depreciations and appreciations, with link to data on futures on the dollar index and its options. The futures on the dollar index, which is composed of the Eurozone euro, Japanese yen, British pound, Canadian dollar, Swedish krona, and Swiss franc, are traded around-the-clock on global platforms. With the U.S. dollar being the reference, the size of the rise of the dollar index futures is shaped by collective concerns about the upside of the dollar index. Our evidence on the size of the option risk premiums with the dollar index futures underlier is at the center of this paper.

A salient aspect of our framework is the ability to examine consistency with prices of options on futures on the dollar index as well as other facts about option and volatility risk premiums, realized (spot and futures) return volatilities, and the dollar index VIX volatility (henceforth labeled $\text{VIX}_t^{\text{dollar index}}$). While existing models are silent about replicating the properties of dollar index option prices, we investigate whether our setup and estimated model can mimic option prices while also meeting empirical regularities.

Collective concerns about U.S. dollar depreciations and appreciations imputed from the options market. We consider the concept of strike-price-dependent dollar concerns, which is based on option risk premiums differentiated by strikes. We find that the more negative risk premiums of calls across higher strike price thresholds quantify investors' dislike for dollar appreciations. Our quantifications about the nature of collective concerns are based on the excess return of options on dollar index futures, rather than on specifications of pricing kernels, the dynamics of futures return volatility, or price jumps.

Using options on futures on the dollar index, we uncover empirical properties that can be used to benchmark model capabilities. We find that the average excess returns of OTM calls on dollar index futures are significantly negative, while average excess returns of OTM puts are insignificant. Additionally, average excess returns of dollar index calls are more negative at higher OTM adjacent strikes, indicating that risk premiums for dollar concerns are strike-price-dependent. This is because the expected payouts of holding OTM calls are lower with declining option deltas, while the purchase costs are relatively higher.

Our findings from options on futures on the dollar index may be confounding to existing models, as they do not consider futures on a currency index and may not agree with data on options on the dollar index futures. Additionally, models may not be able to replicate the size of the negative risk premiums for call options. Furthermore, the dollar index $\mathrm{VIX}_t^{\text{dollar index}}$, which can be derived from prices of options written on futures on the dollar index, has no analytical coverage in existing models of currency behaviors. Options on the dollar index futures provide information not encompassed within individual options on spot currencies.

Empirical modeling of the dollar index futures and options. We develop an estimation framework that focuses on the joint cross-section of dollar index futures and futures return volatilities of the six constituent currencies, which has not been explored previously. We fill a gap by presenting a model of dollar index futures and options, and then compare its predictions against the data. We also examine the agreements with empirical return moments and average excess returns of OTM calls on futures on the dollar index.

The model class we consider provide a closed-form solution for futures on the dollar index, yield semi-analytical expressions for options on the dollar index futures and option risk premiums, and offer an explicit

formula for $\text{VIX}_t^{\text{dollar index}}$. In this model, the futures and option prices are dependent on a variable that drives pricing kernel volatilities.

We estimate the model and undertake statistical inference using data on dollar index futures and futures return variances through quasi-maximum likelihood and Kalman filtering. We assess the model's ability to capture the joint behavior of the cross-section of option prices (differentiated by their strike prices or deltas) and volatilities by taking advantage of its tractability. Error diagnostics show that the model is effective in tracking the time-series evolution of objects of interest, including the seven futures return variances, dollar index option prices, and option volatilities.

The link between the model and data dimensions is explored from several perspectives. Notably, the estimated model can replicate the negative risk premiums of OTM calls on futures on the dollar index. Additionally, the model is consistent with the more negative risk premiums of call options at higher strikes.

We synthesize $\text{VIX}_t^{\text{dollar index}}$ by using a portfolio of options (updated each expiration cycle). We ask: Are model-implied and actual $\text{VIX}_t^{\text{dollar index}}$ close to or far apart from each other? This exercise is important for model assessments because options data was not used in the estimation step. Mincer-Zarnowitz type regressions show that model-implied $\text{VIX}_t^{\text{dollar index}}$ is consistent with its actual counterparts, with a reasonable goodness-of-fit $\overline{R}^2$, and the slope coefficient is positive and statistically significant.

Our consideration of specification error statistics indicates that the model is compatible with data dimensions. Additionally, the estimated model supports dollar index volatility risk premiums that are negative. Our evidence suggests that investors do not like economic states in which the volatility of the pricing kernels is rising.

We establish negative risk premiums linked to low probability events that could drive up the dollar index. This aspect of options data, which is influenced by currency risks across six dimensions, is not adequately assimilated in the literature. Existing currency research has not prioritized the development of empirically performing formulas for futures and options on the geometrically weighted index. We deviate from traditional approaches in our investigation, as outlined in Table 5.

2. Collective concerns about U.S. dollar depreciations or appreciations

By studying options data on futures on the dollar index, we can gain insight into the changing nature of collective concerns about U.S. dollar depreciations or appreciations in the form of state-contingent risk premiums. On the other hand, option prices for individual currencies are influenced by bilateral concerns with the United States, which may not be shared across all constituent economies.

The Intercontinental Exchange (ICE) maintains a (spot) dollar index to monitor the value of the U.S. dollar relative to a currency composite. To outline the ICE convention, let $S_t^{\text{usd}|\text{eur}}$ reflect how many euros convert to one U.S. dollar; that is, the U.S. dollar is the reference. Similar conventions are followed for other exchange rates.

Dollar index construction. The following representation underlies the construction of the dollar index:

$$\begin{aligned}\underbrace{S_t^{\text{usd}|\text{index}}}_{\text{dollar index}} &= 50.14348112\,(S_t^{\text{eur}|\text{usd}})^{-0.576}(S_t^{\text{usd}|\text{jpy}})^{0.136}(S_t^{\text{gbp}|\text{usd}})^{-0.119}(S_t^{\text{usd}|\text{cad}})^{0.091}(S_t^{\text{usd}|\text{sek}})^{0.042}(S_t^{\text{usd}|\text{chf}})^{0.036},\\ &= 50.14348112\,(S_t^{\text{usd}|\text{eur}})^{0.576}(S_t^{\text{usd}|\text{jpy}})^{0.136}(S_t^{\text{usd}|\text{gbp}})^{0.119}(S_t^{\text{usd}|\text{cad}})^{0.091}(S_t^{\text{usd}|\text{sek}})^{0.042}(S_t^{\text{usd}|\text{chf}})^{0.036}. \qquad (1)\end{aligned}$$

The dollar index is the geometric average of the six individual currency pairs: the Eurozone euro, Japanese yen, British pound, Canadian dollar, Swedish krona, and Swiss franc. Furthermore, the dollar index is expressed as the currency composite value per unit of the U.S. dollar. An increase (decrease) in the dollar index is consistent with the appreciation (depreciation) of the U.S. dollar against the currency composite.

In contrast to other dollar indexes, the ICE dollar index has regulated futures and options markets for trading. This market is very liquid according to ICE statistics. Quotes are in U.S. dollars, with Barchart being the source of the daily data on the dollar index, futures, and options on futures on the dollar index.

Futures contract on the dollar index. Let $F_t^{T_F,\text{usd}|\text{index}}$ be the price of the futures written on the dollar index that matures at T_F. A long (short) position in futures entitles the holder to buy (sell) one U.S. dollar in exchange for physically delivering (receiving) the underlying currency components, with weights assigned in equation (1). There may be a case for a negative futures risk premium, as a rise in the dollar index may correspond to undesirable economic conditions.

Options on futures on the dollar index. The data we collect on options includes the strike price, option maturity, and price (quoted in U.S. dollars). We focus on the shortest maturity options, which have an average maturity of 30 days over expiration cycles. In-the-money options are excluded, and our sample starts on 08/03/2001 and ends on 08/07/2020 (229 monthly option expiration cycles). These European-style options are actively traded with an average of 56 puts and calls, with a minimum of 15 strikes for each. The options are physically settled by exchanging the U.S. dollar against the currency composite.

3. Identifying collective concerns from options on the dollar index

Our analysis supports the notion that options on the upside of the dollar index futures (i.e., U.S. dollar appreciations) offer a way to combat undesirable economic uncertainty. The essence of our findings is that investors favor the coverage of OTM calls on dollar index futures across economic states.

3.1. Evidence for negative (insignificant) average excess returns of OTM calls (puts)

Let K be the strike price of the option written on futures on the dollar index. We associate OTM puts with moneyness $k = \log(\frac{K}{F_t^{T_F,\text{usd}|\text{index}}}) < 0$ and OTM calls with $k = \log(\frac{K}{F_t^{T_F,\text{usd}|\text{index}}}) > 0$.

The options expire at time T_O, with $T_O < T_F$. Table 1 (Panel A) reports the average excess returns of options on futures — denominated in U.S. dollars — over expiration cycles, constructed as follows:

$$\mathbb{r}^{\text{put}}_{\{t\to T_O\}}[k] = \frac{\max(K - F_{T_O}^{T_F,\text{usd}|\text{index}}, 0)}{\text{put}_t[K]} - R_t^{\text{us}}, \text{ where } K \text{ corresponds to } \underset{(\text{delta})}{k} = \underset{(-35)}{-1\%}, \underset{(-20)}{-2\%}, \text{ and } \underset{(-10)}{-3\%}, \quad (2)$$

$$\underbrace{\mathbb{r}^{\text{call}}_{\{t\to T_O\}}[k]}_{\substack{\text{in U.S. dollars}\\ \text{(monthly, \%)}}} = \frac{\max(F_{T_O}^{T_F,\text{usd}|\text{index}} - K, 0)}{\text{call}_t[K]} - R_t^{\text{us}}, \text{ where } K \text{ corresponds to } \underset{(\text{delta})}{k} = \underset{(35)}{1\%}, \underset{(20)}{2\%}, \text{ and } \underset{(10)}{3\%}. \quad (3)$$

In equations (2)–(3), $\text{put}_t[K]$ and $\text{call}_t[K]$ represent the time-t prices (in U.S. dollars) for puts and calls with OTM moneyness (absolute delta (as per Black (1976))) closest to 1% (35-delta), 2% (20-delta), and 3% (10-delta). Additionally, $F_{T_O}^{T_F,\text{usd}|\text{index}}$ is the T_O price of the futures contract on the dollar index (with the U.S. dollar as reference), and R_t^{us} is the gross interest rate of the United States.

Table 1 shows the difference in average excess returns between calls and puts of the same moneyness. Notably, the 3% OTM calls exhibit an average excess return of -53%, compared to -13% for puts. These

numbers refer to 30-day average excess returns, and are not annualized. Additionally, the holding period excess returns of 2% OTM calls average -30%, while the corresponding one from puts is positive, at 22%.

How reliable is the pattern that average excess returns of OTM calls are negative? The 95% bootstrap confidence intervals — constructed using a stationary bootstrap procedure — suggest that the average excess returns of 2% and 3% OTM calls are negative and statistically significant. In contrast, the pattern from OTM puts is that their average excess returns are statistically indistinguishable from zero.

Do concerns about dollar appreciation intensify at higher moneyness levels $k > 0$? To answer this question, we investigate if deeper OTM calls are associated with more negative average excess returns. We use a bootstrap to test the following three hypotheses:

$$\underbrace{\overline{\mathbb{r}}^{\text{call}}_{\{t\to T_O\}}[2\%] \;-\; \overline{\mathbb{r}}^{\text{call}}_{\{t\to T_O\}}[1\%]}_{\text{difference in average excess returns}} \geq 0, \quad \overline{\mathbb{r}}^{\text{call}}_{\{t\to T_O\}}[3\%] \;-\; \overline{\mathbb{r}}^{\text{call}}_{\{t\to T_O\}}[2\%] \geq 0, \quad \overline{\mathbb{r}}^{\text{call}}_{\{t\to T_O\}}[3\%] \;-\; \overline{\mathbb{r}}^{\text{call}}_{\{t\to T_O\}}[1\%] \geq 0. \tag{4}$$

Table 2 presents the p-values based on jointly bootstrapping option returns. We assess the empirical likelihood that calls with an OTM moneyness k display higher average returns than the corresponding ones with incrementally lower OTM k. A low p-value indicates rejection of the hypothesis.

Table 2 (Panel A) shows that OTM calls have consistently more negative average returns as $k > 0$ increases. In economic terms, the low p-values from this exercise emphasize that concerns about dollar appreciation intensify at higher $k > 0$, and are thus strike-price-dependent.

Are concerns about the upside of the dollar index more intense, on average, than that of the downside? To address this question using market prices of options, we consider the following hypotheses:

$$\underbrace{\overline{\mathbb{r}}^{\text{call}}_{\{t\to T_O\}}[k]\Big|_{k>0} \;-\; \overline{\mathbb{r}}^{\text{put}}_{\{t\to T_O\}}[k]\Big|_{k<0}}_{\text{difference in average excess returns of calls versus puts}} \;\geq\; 0, \qquad \text{for the same } \; k. \tag{5}$$

The evidence in Table 2 (Panel B) demonstrates that the risk premiums of OTM call options are consistently more negative than that of OTM put options with comparable moneyness and absolute option delta.

We recognize that the expectation of option excess returns under the real-world probability measure reflects option risk premiums. Whereas the average (excess) returns of futures on the dollar index is *insignif-*

icant (Table 1 (Panel B)), our results, and currency index option pricing, are consistent with unconditional risk premiums of OTM calls — quoted closest to 20-delta and 10-delta — that are negative and significant.

3.2. Average excess returns of calls remain negative from the standpoint of non-U.S. investors

Investors based outside of the United States also dislike dollar appreciations. We focus on documenting the excess returns of buying call options (quoted in U.S. dollars) on futures on the dollar index when the trade is initiated from the perspective of non-U.S. investors.

For example, Japanese investors commit to a position, in yen, equivalent to $\text{call}_t[K] \times S_t^{\text{usd}|\text{jpy}}$. Therefore, the excess returns of calls on futures on the dollar index are calculated (in local currency) as follows:

$$\text{Excess returns of calls denominated in yen}_{\{t \to T_O\}}[k] = \frac{S_T^{\text{usd}|\text{jpy}} \times \max(F_{T_O}^{T_F,\text{usd}|\text{index}} - K, 0)}{S_t^{\text{usd}|\text{jpy}} \times \text{call}_t[K]} - R_t^{\text{jpy}}. \tag{6}$$

Table 3 shows that call option risk premiums remain reliably negative and decrease at higher strikes. Hence, buying dollar index calls anchored to alternative currencies does not undercut the idea of strike-price-dependent upside dollar concerns.

3.3. Empirical support from conditional returns of call options on dollar index futures

Realized excess returns reflect ex-ante expected excess returns. To clarify the notation, $\mathbb{E}_t^{\mathbb{P}}(\bullet)$ is a conditional expectation under the real-world measure. To measure the expectation of option returns conditional on the filtration $\mathcal{F}_t$, specifically $\mathbb{E}_t^{\mathbb{P}}(\bullet) = \mathbb{E}^{\mathbb{P}}(\bullet|\mathcal{F}_t)$, we construct average excess option returns conditional on $\{\mathcal{F}_t \in \mathfrak{s}\}$, for some variable $\mathfrak{s}$, over holding periods of option expiration cycles.

To discipline interpretations, each $\mathfrak{s}$ is arranged so as to be in one of the following three groups:

$$\mathcal{F}_t \in \mathfrak{s} = \begin{cases} \mathfrak{s}_{\text{pessimistic}} & \text{(expected excess returns of risky assets is high)}, \\ \mathfrak{s}_{\text{normal}} & \text{(expected excess returns of risky assets is normal)}, \text{ and} \\ \mathfrak{s}_{\text{optimistic}} & \text{(expected excess returns of risky assets is low)}. \end{cases} \tag{7}$$

In order to draw inferences about dollar concerns based on partitioned average excess returns of options, we

focus on two variables to substitute for $\mathfrak{s}$, as follows (known at the beginning of option expiration cycles):

- **Currency volatility** ($\text{Volatility}_t^{\text{currencies}}$). Proxied by JP Morgan currency volatility index (Bloomberg ticker: JPMVXYG7). A high $\text{Volatility}_t^{\text{currencies}}$ encapsulates pessimistic economic states.
- **Change in the price of volatility contract on futures on the dollar index** ($\Delta\text{VOL}_t^{\text{dollar index}}$). The *price* of the volatility contract payoff $\{\log(\frac{F_{T_O}^{T_F,\text{usd|index}}}{F_t^{T_F,\text{usd|index}}})\}^2 = \{-\log(\frac{F_{T_O}^{T_F,\text{index|usd}}}{F_t^{T_F,\text{index|usd}}})\}^2$ is constructed as $\int_0^{F_t^{T_F,\text{usd|index}}} \frac{2(1+\log(\frac{F_t^{T_F,\text{usd|index}}}{K}))}{K^2}\text{put}_t[K]\,dK + \int_{F_t^{T_F,\text{usd|index}}}^{\infty} \frac{2(1-\log(\frac{K}{F_t^{T_F,\text{usd|index}}}))}{K^2}\text{call}_t[K]\,dK$ (employing a standard payoff replication methodology). A high value maps to rising economic uncertainty.

In line with Bakshi, Crosby, Gao, and Hansen (2023), we consider a regression framework in which the excess returns of call options is the dependent variable (likewise for puts), as follows:

$$\mathbb{r}_{\{t\to T_O\}}^{\text{call}}[k] = \underbrace{\eta_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{pessimistic}}\}}^{\text{call}}\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{pessimistic}}\}} + \eta_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{normal}}\}}^{\text{call}}\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{normal}}\}} + \eta_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{optimistic}}\}}^{\text{call}}\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{optimistic}}\}}}_{\text{call option risk premiums on dollar index futures distinguished by economic states}} + \underbrace{\varepsilon_{\{t\to T_O\}}}_{\text{error term}}. \quad (8)$$

For example, $\eta_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{pessimistic}}\}}^{\text{call}}$ reflects the risk premiums of call options in states associated with higher expected returns on risky assets. The presence of $\varepsilon_{\{t\to T_O\}}$ recognizes the distinction between observed excess returns of calls and ex-ante expected excess returns of calls on the dollar index futures. Table 4 shows the estimates of partitioned average excess returns of calls and puts on futures on the dollar index.

While partitioned average excess returns vary with $\{\mathcal{F}_t \in \mathfrak{s}\}$, the common theme is the support for negative conditional risk premiums of call options on futures on the dollar index, which implies that investors pay a cost to protect themselves from possible dollar appreciations in most economic states. Favoring strike-price-dependent upside dollar concerns, the partitioned average excess returns of call options decrease (become more negative) across OTM strikes. Furthermore, the conditional risk premiums of puts are small negative and positive in optimistic states, while they are negative for calls.

In conclusion, we infer collective concerns about dollar appreciations by looking at the call option risk premiums; these become increasingly more negative at higher $k > 0$. By contrast, average excess returns of OTM put options are revealed to be insignificant. Our findings from Tables 1 to 4 on dollar index options are new to the literature. We ask if these findings align with our model of dollar index futures and options.

4. Model of options on futures on the dollar index and empirical consistency

In this section, we analyze the valuation of options on futures on the dollar index, a joint claim on six currencies that make up the index. We explore how models can be differentiated through the sign of the model-implied call option risk premiums across strikes, the capacity to produce plausible option risk premiums, and how they depict option prices on futures on the dollar index.

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0\leq t\leq \mathfrak{T}}, \mathbb{P})$ be a filtered probability space, with $\mathfrak{T}$ being a fixed finite time. The filtration $(\mathcal{F}_t)_{0\leq t\leq \mathfrak{T}}$ satisfies the usual conditions. Let $\mathbb{P}$ denote the real-world probability measure.

Because futures and options on futures on the dollar index are *settled in U.S. dollars*, we focus on a risk-neutral measure $\mathbb{Q}^{\text{us}}$ consistent with the nominal pricing kernel of the United States, denoted M_t^{us}.

Let $\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(\bullet) \equiv \mathbb{E}^{\mathbb{Q}^{\text{us}}}(\bullet|\mathcal{F}_t)$ ($\mathbb{E}_t^{\mathbb{P}}(\bullet) \equiv \mathbb{E}^{\mathbb{P}}(\bullet|\mathcal{F}_t)$) denote expectation under $\mathbb{Q}^{\text{us}}$ ($\mathbb{P}$), conditional on $\mathcal{F}_t$. Furthermore, R_t^{us} (r_t^{us}) is the gross (spot) interest rate of the United States and is assumed to be deterministic.

4.1. Volatility effects do not vanish in the valuation of futures on the dollar index

We explore the interactions among the nominal pricing kernels of the economies that make up the dollar index. To do this, we model a standard Brownian motion $\omega_t^{\mathbb{P}}$ under $\mathbb{P}$, and postulate the following dynamics for the pricing kernels across the seven economies:

$$\underbrace{\frac{dM_t^{\text{us}}}{M_t^{\text{us}}}}_{\text{U.S. pricing kernel}} = -r_t^{\text{us}}\,dt + \alpha_{\text{us}}\sqrt{\mathrm{V}_t}\,d\omega_t^{\mathbb{P}} \quad \text{and} \quad \underbrace{\frac{dM_t^{\text{i}}}{M_t^{\text{i}}}}_{\text{six other pricing kernels}} = -r_t^{\text{i}}\,dt + \alpha_{\text{i}}\sqrt{\mathrm{V}_t}\,d\omega_t^{\mathbb{P}}, \quad \text{i} \in \{1,\ldots,6\}, \tag{9}$$

where α_{us} and α_{i} are constants and r_t^{i} is the spot interest rate in economy i. The variable V_t is the driver of common volatility of the pricing kernel growths.

Volatility variable is mean-reverting, modeled as follows:

$$d\mathrm{V}_t = (\theta_{\text{v}} - \kappa_{\text{v}}\,\mathrm{V}_t)\,dt \;+\; \sigma_{\text{v}}\sqrt{\mathrm{V}_t}\,d\omega_t^{\mathbb{P}}, \qquad \text{where } \theta_{\text{v}}, \kappa_{\text{v}}, \text{ and } \sigma_{\text{v}} \text{ are constants.} \tag{10}$$

We motivate the pricing kernel specifications from the following angles:

- First, constant volatility (a convenient benchmark) has counterfactual implications and constrains

model-based $\mathrm{VIX}_t^{\text{dollar index}}$ to be time-invariant. For identification and ease of model estimation, we consider, but do not parameterize the pairwise correlations among the pricing kernel growths.

- Second, we implement the Mykland and Lee (2008) procedure to detect for jumps using intraday dollar index log futures returns over 30-minute intervals. Out of the 148,647 price observations, only 436 (0.29%) are classified as price jumps.

Our analysis takes into consideration that the dollar index is based upon the U.S. dollar as the reference currency, but is settled in U.S. dollars. In this light, we employ the U.S. dollar as the nonreference in the six currency system, recognizing that modeling of the reciprocal is equivalent with proper measure changes.

The implications of using the U.S. dollar as the nonreference currency in modeling are twofold. First, by Girsanov's theorem, $\omega_t^{\mathbb{Q}^{\text{us}}}$ is a standard Brownian motion under the risk-neutral measure, linked to the real-world measure by $d\omega_t^{\mathbb{P}} = d\omega_t^{\mathbb{Q}^{\text{us}}} + \alpha_{\text{us}}\sqrt{\mathrm{V}_t}\,dt$. Second, the bilateral exchange rates (the U.S. dollar is the nonreference) satisfy $S_t^{\text{i}|\text{usd}} = \frac{M_t^{\text{i}}}{M_t^{\text{us}}}$ for $\text{i} \in \{1,\ldots,6\}$.

By Ito's lemma, $\frac{dS_t^{\text{i}|\text{usd}}}{S_t^{\text{i}|\text{usd}}} = \frac{dM_t^{\text{i}}}{M_t^{\text{i}}} - \frac{dM_t^{\text{us}}}{M_t^{\text{us}}} + \{\frac{dM_t^{\text{us}}}{M_t^{\text{us}}}\}^2 - \frac{dM_t^{\text{us}}}{M_t^{\text{us}}} \times \frac{dM_t^{\text{i}}}{M_t^{\text{i}}}$. Disciplining constructions, the dynamics of the six exchange rates, with the U.S. dollar as the *nonreference* currency, are as follows:

$$\underbrace{\frac{dS_t^{\text{i}|\text{usd}}}{S_t^{\text{i}|\text{usd}}}}_{\text{usd is nonreference}} = \begin{cases} (r_t^{\text{us}} - r_t^{\text{i}})\,dt \;+\; \alpha_{\text{us}}(\alpha_{\text{us}} - \alpha_{\text{i}})\,\mathrm{V}_t\,dt \;+\; \{\alpha_{\text{i}} - \alpha_{\text{us}}\}\sqrt{\mathrm{V}_t}\,d\omega_t^{\mathbb{P}}, & (\text{under } \mathbb{P}) \\ (r_t^{\text{us}} - r_t^{\text{i}})\,dt \;+\; \{\alpha_{\text{i}} - \alpha_{\text{us}}\}\sqrt{\mathrm{V}_t}\,d\omega_t^{\mathbb{Q}^{\text{us}}}. & (\text{under } \mathbb{Q}^{\text{us}}) \end{cases} \tag{11}$$

Consistent with our objectives, we build currency index mechanics that bundle currency risks, as follows:

$$\underbrace{S_t^{\text{index}|\text{usd}}}_{\text{usd is nonreference}} = \frac{1}{\underbrace{S_t^{\text{usd}|\text{ index}}}_{\text{usd is reference}}} = \frac{1}{50.14348112 \times \prod_{\text{i}=1}^{6}(S_t^{\text{usd}|\text{i}})^{\mathbb{w}_{\text{i}}}} = \frac{1}{50.14348112}\prod_{\text{i}=1}^{6}(S_t^{\text{i}|\text{usd}})^{\mathbb{w}_{\text{i}}}, \tag{12}$$

for weights $\mathbb{w}_{\text{i}}$, indexed by $\text{i} \in \{1,\ldots,6\}$, with $\sum_{\text{i}} \mathbb{w}_{\text{i}} = 1$, satisfying the following composition:

$$\mathbb{w}_1 = 0.576, \quad \mathbb{w}_2 = 0.136, \quad \mathbb{w}_3 = 0.119, \quad \mathbb{w}_4 = 0.091, \quad \mathbb{w}_5 = 0.042, \quad \text{and} \quad \mathbb{w}_6 = 0.036. \tag{13}$$

In the manner of its construction, write $\underbrace{S_t^{\text{index}|\text{usd}}[S_t^{1|\text{usd}},\ldots,S_t^{6|\text{usd}}]}_{\text{contingent claim on six currencies}}$. An application of Ito's lemma yields as follows:

$$\underbrace{\frac{dS_t^{\text{index}|\text{usd}}}{S_t^{\text{index}|\text{usd}}}}_{\substack{\text{usd is}\\ \text{nonreference}}} = \begin{cases} (r_t^{\text{us}} - r_t^{\text{index}})\,dt + \alpha_{\text{us}}\{\alpha_{\text{us}} - \sum_{\text{i}} \mathbb{w}_{\text{i}}\,\alpha_{\text{i}}\}\,\text{V}_t\,dt + \frac{1}{2}\sum_{\text{i}}(\mathbb{w}_{\text{i}})(\mathbb{w}_{\text{i}}-1)\,\{\alpha_{\text{i}} - \alpha_{\text{us}}\}^2\,\text{V}_t dt \\ + \frac{1}{2}\sum_i \sum_{j\neq i} 2\mathbb{w}_{\text{i}}\,\mathbb{w}_{\text{j}} \times \{\alpha_{\text{i}} - \alpha_{\text{us}}\}\{\alpha_{\text{j}} - \alpha_{\text{us}}\}\text{V}_t\,dt + \sum_{\text{i}} \mathbb{w}_{\text{i}}\,\{\alpha_{\text{i}} - \alpha_{\text{us}}\}\sqrt{\text{V}_t}\,d\omega_t^{\mathbb{P}}, \quad \leftarrow \text{under } \mathbb{P} \\ (r_t^{\text{us}} - r_t^{\text{index}})\,dt + \frac{1}{2}\sum_{\text{i}}(\mathbb{w}_{\text{i}})(\mathbb{w}_{\text{i}}-1)\,\{\alpha_{\text{i}} - \alpha_{\text{us}}\}^2\,\text{V}_t dt \\ + \frac{1}{2}\sum_i \sum_{j\neq i} 2\mathbb{w}_{\text{i}}\,\mathbb{w}_{\text{j}} \times \{\alpha_{\text{i}} - \alpha_{\text{us}}\}\{\alpha_{\text{j}} - \alpha_{\text{us}}\}\text{V}_t\,dt + \sum_{\text{i}} \mathbb{w}_{\text{i}}\,\{\alpha_{\text{i}} - \alpha_{\text{us}}\}\sqrt{\text{V}_t}\,d\omega_t^{\mathbb{Q}^{\text{us}}}, \quad \leftarrow \text{under } \mathbb{Q}^{\text{us}} \end{cases} \tag{14}$$

where we define $r_t^{\text{index}} \equiv \sum_{\text{i}} \mathbb{w}_{\text{i}} r_t^{\text{i}}$.

A departure — not necessarily conveyed in the workings of extant models — is that the instantaneous drift of $\frac{dS_t^{\text{index}|\text{usd}}}{S_t^{\text{index}|\text{usd}}}$ under $\mathbb{Q}^{\text{us}}$ deviates from $r_t^{\text{us}} - r_t^{\text{index}}$ and is connected to V_t.

Appendix A shows that the dollar index futures price, determined by solving $\frac{F_t^{T_F,\text{index}|\text{usd}}}{S_t^{\text{index}|\text{usd}}} = \mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(\frac{S_{T_F}^{\text{index}|\text{usd}}}{S_t^{\text{index}|\text{usd}}})$, is linked to the volatility of pricing kernel growths, for futures maturity $\tau \equiv T_F - t$, as follows:

$$\overbrace{\frac{S_t^{\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}}^{\text{usd is reference}} = \overbrace{\frac{F_t^{T_F,\text{index}|\text{usd}}}{S_t^{\text{index}|\text{usd}}}}^{\text{usd is nonreference}} = \exp\Big((r_t^{\text{us}} - r_t^{\text{index}})\,\tau + \mathbb{a}[\tau] + \mathbb{b}[\tau]\,\text{V}_t\Big). \tag{15}$$

We determine that

$$\mathbb{b}[\tau] = \frac{2(A + \frac{1}{2}B^2)(1 - e^{-\mho\tau})}{2\mho - (\mho - (\kappa_{\text{v}} - \alpha_{\text{us}}\sigma_{\text{v}} - \sigma_{\text{v}}B))(1 - e^{-\mho\tau})}, \tag{16}$$

$$\mathbb{a}[\tau] = -\frac{\theta_{\text{v}}}{\sigma_{\text{v}}^2}\Big((\mho - (\kappa_{\text{v}} - \alpha_{\text{us}}\sigma_{\text{v}} - \sigma_{\text{v}}B))\,\tau + 2\log(1 - \frac{(\mho - (\kappa_{\text{v}} - \alpha_{\text{us}}\sigma_{\text{v}} - \sigma_{\text{v}}B))(1 - e^{-\mho\tau})}{2\mho})\Big), \tag{17}$$

$$\text{where } \mho \equiv \sqrt{(\kappa_{\text{v}} - \alpha_{\text{us}}\sigma_{\text{v}} - \sigma_{\text{v}}B)^2 - 2\sigma_{\text{v}}^2(A + \frac{1}{2}B^2)} \quad \text{with} \quad B \equiv \sum_{\text{i}} \mathbb{w}_{\text{i}}\,\{\alpha_{\text{i}} - \alpha_{\text{us}}\}, \quad \text{and} \tag{18}$$

$$A \equiv \frac{1}{2}\sum_{\text{i}}(\mathbb{w}_{\text{i}})(\mathbb{w}_{\text{i}}-1)\,\{\alpha_{\text{i}} - \alpha_{\text{us}}\}^2 + \frac{1}{2}\sum_i\sum_{j\neq i}(2\mathbb{w}_{\text{i}}\,\mathbb{w}_{\text{j}})\,\{\alpha_{\text{i}} - \alpha_{\text{us}}\}\{\alpha_{\text{j}} - \alpha_{\text{us}}\} - \frac{1}{2}\Big\{\sum_{\text{i}} \mathbb{w}_{\text{i}}\,\{\alpha_{\text{i}} - \alpha_{\text{us}}\}\Big\}^2. \tag{19}$$

Contrasting $\frac{F_t^{T_F,\text{i}|\text{usd}}}{S_t^{\text{i}|\text{usd}}} = \mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(\frac{S_{T_F}^{\text{i}|\text{usd}}}{S_t^{\text{i}|\text{usd}}}) = e^{(r_t^{\text{us}} - r_t^{\text{i}})(T_F - t)}$, the novel aspect of (15) is that a higher $F_t^{T_F,\text{usd}|\text{index}}$ is associated with a higher V_t (i.e., $\mathbb{b}[\tau] < 0$). Inviting an explanation, this *inverse leverage* effect arises because dollar risks are retained in the geometrically weighted index construction. This finding is pertinent to our evidence, as dollar index futures can be considered to mimic the properties of a safe haven security.

The structure of pricing equation (15) facilitates quasi-maximum likelihood estimation in combination with Kalman filtering. Ensuring consistency of relationships, $(F_\ell^{\text{index}|\text{usd}})$ is a $\mathbb{Q}^{\text{us}}$ martingale, as follows:

$$\underbrace{\frac{dF_t^{T_F,\text{index}|\text{usd}}}{F_t^{T_F,\text{index}|\text{usd}}}}_{\text{usd is nonreference}} = \begin{cases} -\sigma_F[T_F-t]\,\alpha_{\text{us}}\,\text{V}_t\,dt \;+\; \sigma_F[T_F-t]\sqrt{\text{V}_t}\,d\omega_t^{\mathbb{P}}, & \leftarrow \text{under } \mathbb{P} \\ \sigma_F[T_F-t]\sqrt{\text{V}_t}\,d\omega_t^{\mathbb{Q}^{\text{us}}}, & \leftarrow \text{under } \mathbb{Q}^{\text{us}} \\ \quad \text{where it holds that } \sigma_F[T_F-t] \;\equiv \underbrace{\mathbb{b}[T_F-t]}_{\substack{\text{deterministic} \\ \text{(equation (16))}}}\sigma_{\text{V}} + \sum_{\text{i}} \mathbb{w}_{\text{i}}(\alpha_{\text{i}}-\alpha_{\text{us}}). \end{cases} \tag{20}$$

One may determine the dynamics $\frac{dF_t^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}} = -\frac{dF_t^{T_F,\text{index}|\text{usd}}}{F_t^{T_F,\text{index}|\text{usd}}} + \{\frac{dF_t^{T_F,\text{index}|\text{usd}}}{F_t^{T_F,\text{index}|\text{usd}}}\}^2$. In this model, the (conditional) dollar index futures risk premium is time-varying and impacted by V_t.

4.2. Risk premiums of options on futures on the dollar index

Emphasizing that the dollar index is quoted with U.S. dollar as the reference and settles in U.S. dollars, we write $K^{\text{usd}|\text{index}} > F_t^{T_F,\text{usd}|\text{index}}$ as the strike price of the OTM call on dollar index futures. The moneyness is $k^{\text{usd}|\text{index}} = \log(\frac{K^{\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}) > 0$. Rearranging reciprocal relationships, $k^{\text{index}|\text{usd}} = \log(\frac{K^{\text{index}|\text{usd}}}{F_t^{T_F,\text{index}|\text{usd}}}) < 0$. We denote the log futures return — with U.S. dollar as the nonreference currency — by $\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}} \equiv \log(\frac{F_{T_O}^{T_F,\text{index}|\text{usd}}}{F_t^{T_F,\text{index}|\text{usd}}})$.

In what follows, $i=\sqrt{-1}$ and $\text{Re}[\bullet]$ are the real part of the complex-valued function. Making use of an exponential Fourier transform identity (Bateman and Erdelyi (1954, 3.2.3, page 118)), Appendix B shows that the risk premium of call option on the dollar index futures can be written as follows:

$$\text{Call risk premium}_{\{t\to T_O\}}[K^{\text{usd}|\text{index}}] \;\approx\; R_t^{\text{us}}\left(\frac{\frac{1}{2\pi}\int_{-\infty}^{\infty}\text{Re}[\frac{e^{\{1+i\phi\}k^{\text{index}|\text{usd}}}}{(1+i\phi)^2}\,\mathbb{E}_t^{\mathbb{P}}(e^{-\{1+i\phi\}\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}}})]d\phi}{\frac{1}{2\pi}\int_{-\infty}^{\infty}\text{Re}[\frac{e^{\{1+i\phi\}k^{\text{index}|\text{usd}}}}{(1+i\phi)^2}\,\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\{1+i\phi\}\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}}})]d\phi} \;-\; 1\right). \tag{21}$$

The relevance of the (approximate) treatment in (21) is that the negative risk premium of OTM calls — with strike $K^{\text{usd}|\text{index}} > F_t^{T_F,\text{usd}|\text{index}}$ — can be shown to be restricted by risk premiums on powers of log futures returns (see (B8) of Appendix B). In economic terms, the risk premium of OTM calls can be negative, provided the risk-neutral measure dominates the real-world measure as the U.S. dollar is appreciating.

For empirical implementations, the expressions for $\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\{1+i\phi\}\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}}})$ and $\mathbb{E}_t^{\mathbb{P}}(e^{-\{1+i\phi\}\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}}})$ — specific to our model in (20) — are shown in (B12) and (B13) of Appendix B, respectively. Notably, the option risk premiums vary with volatility variable V_t in the model. We now turn to establishing the quantitative relevance of the estimated model in (20) and the exactly formulated option pricing model.

5. Evaluating the quantitative implications of the model

Despite the substantial work on exchange rates, we are unaware of prior research — see the list in Table 5 — that explores the relationship of models to data on options on futures on the dollar index. We consider a model for options on futures on the dollar index with two advantages: (i) the OTM call option risk premiums can be negative, and (ii) the option prices can be expressed semi-analytically. For comparison, it is unclear whether extant models can be modified to distill either feature (i) or (ii), or both.

5.1. Implications of model parameter estimates across the system of seven economies

We investigate the model's ability to capture the dynamics of futures on the dollar index, options on futures, and volatilities in the cross-section of individual currency futures. To estimate model parameters and extract $\{\mathrm{V}_t : t = 1, \ldots, T\}$, we employ the Kalman filter and quasi-maximum likelihood.

Suppose $\mathbf{rv}_t^{\text{futures}}$ is the 6×1 vector of annualized realized variances constructed from daily log futures returns over option expiration cycles. Aligning with dynamics (11) and mapping to *individual* log currency futures returns (i.e., consistent with $\frac{F_t^{T_F,\text{i}|\text{usd}}}{S_t^{\text{i}|\text{usd}}} = e^{(r_t^{\text{us}} - r_t^{\text{i}})(T_F - t)}$), the (annualized) expected integrated variance over t and $t + \Delta$ is

$$\frac{1}{\Delta}\mathbb{E}_t^{\mathbb{P}}\Big(\int_t^{t+\Delta}\Big\{\frac{1}{du}\text{var}_t^{\mathbb{P}}[d\log(F_u^{T_F,\text{i}|\text{usd}})]\Big\}\,du \;=\; \mathbb{c}[\Delta] \;+\; \mathbb{d}[\Delta]\,\mathrm{V}_t, \tag{22}$$

where $\mathbb{c}[\Delta] = \{\alpha_{\text{i}} - \alpha_{\text{us}}\}^2 \frac{\theta_{\text{v}}}{\kappa_{\text{v}}}\{1 - \frac{(1-e^{-\kappa_{\text{v}}\Delta})}{\kappa_{\text{v}}\Delta}\}$, $\mathbb{d}[\Delta] = \{\alpha_{\text{i}} - \alpha_{\text{us}}\}^2 \frac{(1-e^{-\kappa_{\text{v}}\Delta})}{\kappa_{\text{v}}\Delta}$, and $\Delta = \frac{30}{365}$.

In light of (15), the measurement equations — that identify all parameters — take the following form:

$$\underbrace{\begin{pmatrix} \frac{1}{\tau}\log\Big(\frac{F_t^{T_F,\text{usd}|\text{index}}\, e^{-(r_t^{\text{index}} - r_t^{\text{us}})\tau}}{S_t^{\text{usd}|\text{index}}}\Big) \\ \underset{(6\times 1)}{\mathbf{rv}_t^{\text{futures}}} \end{pmatrix}}_{\mathbb{y}_t \;\text{(data)}} = \underbrace{\begin{pmatrix} -\frac{\mathbb{a}[\tau]}{\tau} - \frac{\mathbb{b}[\tau]}{\tau}\mathrm{V}_t \\ \mathbb{c}[\Delta] + \mathbb{d}[\Delta]\,\mathrm{V}_t \end{pmatrix}}_{\mathbb{h}[\mathrm{V}_t;\Theta]} + \begin{pmatrix} \sigma^{\text{f}} I_1 & 0 \\ 0 & \sigma^{\text{rv}} \underbrace{I_6}_{\substack{\text{identity}\\ \text{matrix}}} \end{pmatrix} \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim N(\mathbf{0}, \Omega). \tag{23}$$

The measurement equations depict the relation between V_t and (i) the dollar index futures prices and (ii) the six futures return variances. In (23), $\mathbb{h}[\mathrm{V}_t;\Theta]$ is formalized by the model, $\boldsymbol{\varepsilon}_t$ reflects IID Gaussian errors with covariance matrix Ω, and $\underset{17\times 1}{\Theta} = [\theta_{\text{v}}, \kappa_{\text{v}}, \sigma_{\text{v}}, \alpha_{\text{us}}, \alpha_{\text{eurozone}}, \ldots, \alpha_{\text{switzerland}}, \sigma^{\text{f}}, \sigma^{\text{rv}}_{\text{eurozone}}, \ldots, \sigma^{\text{rv}}_{\text{switzerland}}]$ is the vector of parameters to be estimated. The individual futures return variances identify α_{i}, whereas the

dollar index futures prices isolate the risk premiums parameterized by α_{us}. The variance risks are priced.

The transition equation depicts the discrete-time evolution of V_t. As per tradition, we approximate the transition density of V_t with a Gaussian density with the same first and second moments (e.g., Carr and Wu (2007) and Jacobs and Karoui (2009)). We note that

$$\mathrm{V}_t \;=\; \frac{\theta_{\text{v}}}{\kappa_{\text{v}}}(1-e^{-\kappa_{\text{v}}\Delta}) \;+\; e^{-\kappa_{\text{v}}\Delta}\mathrm{V}_{t-1} \;+\; \eta_t, \qquad \eta_t \sim \mathcal{N}(0,\mathcal{V}_t), \qquad \text{and} \qquad \Delta = \frac{30}{365}, \tag{24}$$

where $\mathcal{V}_t = \frac{\sigma_{\text{v}}^2(1-e^{-\kappa_{\text{v}}\Delta})}{\kappa_{\text{v}}}\{\frac{1}{2}\frac{\theta_{\text{v}}}{\kappa_{\text{v}}}(1-e^{-\kappa_{\text{v}}\Delta})+e^{-\kappa_{\text{v}}\Delta}\mathrm{V}_{t-1}\}$ and $\mathbb{E}^{\mathbb{P}}_{t-1}(V_t) = \frac{\theta_{\text{v}}}{\kappa_{\text{v}}}(1-e^{-\kappa_{\text{v}}\Delta})+e^{-\kappa_{\text{v}}\Delta}\mathrm{V}_{t-1}$.

The Kalman filter is associated with one-step-ahead forecasts for $\mathbb{y}_t$, denoted $\widehat{\mathbb{y}}_{t|t-1}$, and the corresponding error covariance matrix, $\Sigma_{t|t-1}$. We construct the log-likelihood function as follows:

$$\mathfrak{L}[\Theta] \;=\; -\frac{1}{2}\sum_{t=1}^{T}\{\#\log(2\pi) \;+\; \log\left|\Sigma_{t|t-1}\right| \;+\; (\mathbb{y}_{t|t-1}-\widehat{\mathbb{y}}_{t|t-1})^{\top}\Sigma_{t|t-1}^{-1}(\mathbb{y}_{t|t-1}-\widehat{\mathbb{y}}_{t|t-1})\}, \tag{25}$$

where T is the number of option expiration cycles. The quasi-maximum likelihood estimator of Θ is

$$\hat{\Theta} \;=\; \arg\max_{\Theta} \;\; \mathfrak{L}[\Theta]. \tag{26}$$

Table 6 reports the parameters and features of the extracted $\{V_t : t = 1,\ldots,T\}$ used in model implementation. The displayed asymptotic standard errors are obtained from the Fisher information matrix. We focus on questions related to option risk premiums, quantitative model assessments, and statistical significance.

Although volatility has been estimated in different markets, note that the reported θ_{v}, κ_{v}, and σ_{v} impact the properties of each pricing kernel's volatility. For example, the reported $\theta_{\text{v}} = 0.906$ and $\kappa_{\text{v}} = 3.004$ correspond to an annualized long run $\sqrt{\mathrm{V}_t}$ of 54.9%=$\sqrt{\frac{0.906}{3.004}}$ (Table 6 (Panel A)). These parameters differ, for instance, from Bates (1996, Table 2), since he studies the risk-neutral process of the Deutsche mark and not pricing kernels.

Similarly, the estimate of $\sigma_{\text{v}} = 1.192$ deviates from its counterparts in equity index studies. This is because pricing kernel volatility is much higher than equity index volatility.

When the model is estimated by combining the time-series and cross-sectional information anchored to theory, the reported $\alpha_{\text{us}} = 1.268$, $\alpha_{\text{eurozone}} = 1.098$, $\alpha_{\text{japan}} = 1.101$, $\alpha_{\text{uk}} = 1.097$, $\alpha_{\text{canada}} = 1.098$, $\alpha_{\text{sweden}} = 1.046$,

and $\alpha_{\text{switzerland}} = 1.085$ support heterogeneity in volatility of pricing kernel growths. These estimates imply long-run pricing kernel volatilities between 57.4% and 69.6% annualized, evidence that also agrees with the treatment of Brandt, Cochrane, and Santa-Clara (2006). Figure 1 plots the time-series of extracted $\sqrt{\mathrm{V}_t}$.

5.2. How well does the model capture the behavior of option prices?

Drawing on the extracted V_t and model parameters, we next examine the quantitative fit to call and put option prices. The model price of options on futures on the dollar index are derived as follows:

$$\underbrace{\text{call}_t[K^{\text{usd}|\text{index}}]}_{\text{usd is reference}} = \frac{F_t^{T_F,\text{usd}|\text{index}}}{R_t^{\text{us}}}\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\mathbb{x}^{\text{index}|\text{usd}}_{\{t\to T_O\}}}) \times \{\frac{1}{2} - \frac{1}{\pi}\int_0^\infty \text{Re}[\frac{e^{-i\phi k^{\text{index}|\text{usd}}}\frac{\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{\{i\phi-1\}\mathbb{x}^{\text{index}|\text{usd}}})}{\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\mathbb{x}^{\text{index}|\text{usd}}})}}{i\phi}]\,d\phi\}$$
$$- \frac{K^{\text{usd}|\text{index}}}{R_t^{\text{us}}} \times \{\frac{1}{2} - \frac{1}{\pi}\int_0^\infty \text{Re}[\frac{e^{-i\phi k^{\text{index}|\text{usd}}}\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{i\phi\mathbb{x}^{\text{index}|\text{usd}}})}{i\phi}]\,d\phi\}, \qquad \text{and} \tag{27}$$

$$\underbrace{\text{put}_t[K^{\text{usd}|\text{index}}]}_{\text{usd is reference}} = \frac{K^{\text{usd}|\text{index}}}{R_t^{\text{us}}}\{\frac{1}{2} + \frac{1}{\pi}\int_0^\infty \text{Re}[\frac{e^{-i\phi k^{\text{index}|\text{usd}}}\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{i\phi\mathbb{x}^{\text{index}|\text{usd}}})}{i\phi}]d\phi\}$$
$$- \frac{F_t^{T_F,\text{usd}|\text{index}}}{R_t^{\text{us}}}\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\mathbb{x}^{\text{index}|\text{usd}}})\{\frac{1}{2} + \frac{1}{\pi}\int_0^\infty \text{Re}[\frac{e^{-i\phi k^{\text{index}|\text{usd}}}\frac{\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{\{i\phi-1\}\mathbb{x}^{\text{index}|\text{usd}}})}{\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\mathbb{x}^{\text{index}|\text{usd}}})}}{i\phi}]d\phi\}. \tag{28}$$

The parametric forms of $\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{\{i\phi-1\}\mathbb{x}^{\text{index}|\text{usd}}_{\{t\to T_O\}}})$, $\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{i\phi\mathbb{x}^{\text{index}|\text{usd}}_{\{t\to T_O\}}})$, and $\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\mathbb{x}^{\text{index}|\text{usd}}_{\{t\to T_O\}}})$ for option pricing are obtained from equation (B12) of Appendix B. To our knowledge, the model presented in this paper is the first model for semi-analytically pricing options on futures on the dollar index.

Table 7 examines the model's ability to price options by looking at average model errors and the absolute model errors. The model errors are calculated by comparing the model-implied option price to the actual price, and by comparing the $\{\sigma_t^{\text{Black}}\}^{\text{model}}$ to the $\{\sigma_t^{\text{Black}}\}^{\text{actual}}$, where σ_t^{Black} is inferred from Black (1976).

The key finding from Table 7 is that our model provides a reasonable reconciliation of the data. For example, the average absolute errors for 3% OTM puts (calls) are 0.077(0.121). The slope coefficients in the Mincer-Zarnowitz regressions with $\log(\{\sigma_t^{\text{Black}}\}^{\text{actual}})$ as the dependent variable range between 0.84 and 0.85 (0.42 and 0.76) for calls (puts). The $\overline{R}^2$'s values from the regressions range from 59% to 69%. It bears

emphasizing that measurement equations in the estimation procedure do not include option prices.

5.3. The model can produce negative risk premiums of both OTM calls and return volatility

To replicate the environment of Table 1, we apply the formula below:

$$\text{Call risk premium}_{\{t\to T_O\}}[K^{\text{usd}|\text{index}}] \;\equiv\; \frac{\mathbb{E}_t^{\mathbb{P}}(\max(F_{T_O}^{T_F,\text{usd}|\text{index}} - K^{\text{usd}|\text{index}},0))}{\text{call}_t[K^{\text{usd}|\text{index}}]} \;-\; R_t^{\text{us}}. \tag{29}$$

The expression for $\mathbb{E}_t^{\mathbb{P}}(\max(F_{T_O}^{T_F,\text{usd}|\text{index}} - K^{\text{usd}|\text{index}},0))$ is shown in (B16), whereas the model call option price is in (27). We compute the numerator and denominator (29) at the beginning of option expiration cycles while matching the option maturity and moneyness. Utilizing the estimated model parameters and extracted V_t, we report the OTM call option risk premiums in Table 8.

Connecting to our findings, two implications emerge. Firstly, qualitatively matching the patterns in Table 1, the OTM call option risk premiums from the model are negative. Secondly, as in Table 2, the call risk premiums are reliably more negative deeper OTM (the NW[p] for this test is 0.000). Therefore, the imputed call option risk premiums appear to be consistent with time-varying collective concerns about dollar appreciations.

Furthermore, our analysis leads us to study the log volatility risk premium as:

$$\text{Volatility risk premium}_{\{t\to T_O\}} \;\equiv\; \log\Big(\sqrt{\frac{1}{T_O-t}\text{var}_t^{\mathbb{P}}[\frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}]}\Big/\sqrt{\frac{1}{T_o-t}\text{var}_t^{\mathbb{Q}^{\text{us}}}[\frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}]}\Big). \tag{30}$$

Equations (C4) and (C7) of Appendix C analytically compute $\text{var}_t^{\mathbb{Q}^{\text{us}}}[\frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}]$ and $\text{var}_t^{\mathbb{P}}[\frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}]$ in terms of estimated model parameters and extracted V_t.

The average of estimated model volatility risk premiums is -9.4%, as seen in Table 8. This magnitude brackets the bootstrap confidence intervals of volatility risk premiums imputed from the data, which is $\lfloor -13.9\%, -7.2\% \rfloor$. Our analysis indicates that investors are willing to pay a premium to cover dollar index volatility risks.

In sum, the estimated model implies that OTM call option risk premiums and volatility risk premiums

are both negative.

5.4. Implications of extracted $\sqrt{\mathrm{V}_t}$ for the dynamics of $\mathrm{VIX}_t^{\mathrm{usd}|\mathrm{index}}$

Our framework connects to the data on a portfolio of options on futures on the dollar index by formulating $\{\mathrm{VIX}_t^{\mathrm{usd}|\mathrm{index}}\}^2$, an object not emphasized in estimated currency models. Consistent with the convexity of $\{\frac{-2}{T_O-t}\}\log(\frac{F_{T_O}^{T_F,\mathrm{index}|\mathrm{usd}}}{F_t^{T_F,\mathrm{index}|\mathrm{usd}}})$ in $F_{T_O}^{T_F,\mathrm{index}|\mathrm{usd}}$, we obtain as follows:

$$
\begin{aligned}
\{\mathrm{VIX}_t^{\mathrm{usd}|\mathrm{index}}\}^2 \quad &\equiv \quad \mathbb{E}_t^{\mathbb{Q}^{\mathrm{us}}}(\{\frac{-2}{T_O-t}\}\log(\frac{F_{T_O}^{T_F,\mathrm{index}|\mathrm{usd}}}{F_t^{T_F,\mathrm{index}|\mathrm{usd}}})), \\
&\underbrace{=}_{\textbf{model}} \quad \frac{\{\mathbb{b}[\tau]\,\sigma_{\mathrm{v}} \,+\, \sum_{\mathrm{i}} \mathrm{w}_{\mathrm{i}}(\alpha_{\mathrm{i}}-\alpha_{\mathrm{us}})\}^2}{T_O-t}\Big(\frac{(T_O-t)\,\theta_{\mathrm{v}}}{\kappa_{\mathrm{v}}-\alpha_{\mathrm{us}}\,\sigma_{\mathrm{v}}} - \frac{(1-e^{-\{\kappa_{\mathrm{v}}-\alpha_{\mathrm{us}}\,\sigma_{\mathrm{v}}\}\,(T_O-t)})}{\kappa_{\mathrm{v}}-\alpha_{\mathrm{us}}\,\sigma_{\mathrm{v}}}\{\frac{\theta_{\mathrm{v}}}{\kappa_{\mathrm{v}}-\alpha_{\mathrm{us}}\,\sigma_{\mathrm{v}}}\}\Big) \\
&\qquad + \left(\frac{\{\mathbb{b}[\tau]\,\sigma_{\mathrm{v}} \,+\, \sum_{\mathrm{i}} \mathrm{w}_{\mathrm{i}}(\alpha_{\mathrm{i}}-\alpha_{\mathrm{us}})\}^2}{T_O-t} \times \frac{(1-e^{-\{\kappa_{\mathrm{v}}-\alpha_{\mathrm{us}}\,\sigma_{\mathrm{v}}\}\,(T_O-t)})}{\kappa_{\mathrm{v}}-\alpha_{\mathrm{us}}\,\sigma_{\mathrm{v}}}\right)\mathrm{V}_t, \qquad (31) \\
&\underbrace{=}_{\textbf{data}} \quad \frac{2R_t^{\mathrm{us}}}{T_O-t}\left(\int_0^{F_t^{T_F,\mathrm{usd}|\mathrm{index}}} \frac{\mathrm{put}_t[K]}{K^2}\,dK \,+\, \int_{F_t^{T_F,\mathrm{usd}|\mathrm{index}}}^{\infty} \frac{\mathrm{call}_t[K]}{K^2}\,dK\right). \qquad (32)
\end{aligned}
$$

Table 9 (Panels A and B) tabulates the properties of model-derived $\{\mathrm{VIX}_t^{\mathrm{usd}|\mathrm{index}}\}^2$ from equation (31) and option-inferred $\{\mathrm{VIX}_t^{\mathrm{usd}|\mathrm{index}}\}^2$ from equation (32). To evaluate the model performance, we compare the model-inferred and option-inferred $\mathrm{VIX}_t^{\mathrm{usd}|\mathrm{index}}$, focusing on option portfolios represented by this quantity which is not a part of the measurement equations.

The properties of model-inferred and actual $\mathrm{VIX}_t^{\mathrm{usd}|\mathrm{index}}$ appear to be aligned in levels across the various percentiles. We can gain further insights by performing Mincer-Zarnowitz type regressions:

$$
\log(\{\mathrm{VIX}_t^{\mathrm{usd}|\mathrm{index}}\}^{\mathrm{actual}}) \;=\; \beta_0 \,+\, \beta_1\,\log(\{\mathrm{VIX}_t^{\mathrm{usd}|\mathrm{index}}\}^{\mathrm{model}}) \,+\, e_t. \qquad (33)
$$

Table 10 (Panel A) shows a statistically significant slope coefficient β_1 of 0.68 and an $\overline{R}^2$ of 53%. This indicates that the model time-series has information for actual $\mathrm{VIX}_t^{\mathrm{usd}|\mathrm{index}}$. Additionally, the two series display a tendency to comove together with a correlation of 0.73.

Table 9 (Panel B) presents the contemporaneous correlations between seven futures returns and changes in actual and model-implied $\mathrm{VIX}_t^{\mathrm{usd}|\mathrm{index}}$. For instance, the correlation between changes in $\log(\{\mathrm{VIX}_t^{\mathrm{usd}|\mathrm{index}}\}^{\mathrm{actual}})$

and $\mathbb{r}^{\text{futures,usd}|\text{index}}_{\{t\to T_O\}}$ is 0.20, while the corresponding one for $\log(\{\text{VIX}_t^{\text{usd}|\text{index}}\}^{\text{model}})$ is 0.21. These results demonstrate the time-series and cross-market aspects of the model.

To analyze $\text{VIX}_t^{\text{usd}|\text{index}}$, we employ two measures of model goodness of fit: The average errors and absolute errors. We compute the model-determined quantity — based on estimated model parameters and the extracted V_t — and compare to the actual, to obtain model errors. Updating the procedure over each option expiration cycle, we tabulate the error metrics in Table 10 (Panel A). The average gap between model and actual $\text{VIX}_t^{\text{usd}|\text{index}}$ is 54 bps, and the average absolute errors are 141 bps.

5.5. Additional yardsticks of model adequacy and average cross-sectional $\overline{R}^2$

Following (20), we deduce the expected integrated variance of dollar index futures returns as $\frac{1}{\Delta}\int_t^{t+\Delta}\sigma_F^2[T_F - t]\{\frac{\theta_\text{v}}{\kappa_\text{v}} + (\text{V}_t - \frac{\theta_\text{v}}{\kappa_\text{v}})e^{-\kappa_\text{v}(u-t)}\}\,du$. Note that this quantity — which has a closed-form — is not a part of measurement equations. The performance results — shown in Table 10 (Panel A) — imply average (absolute) errors of 143 (148) bps. The Mincer-Zarnowitz type regressions of conformity indicate an $\overline{R}^2$ of 86%.

In addition, we assess the model performance by fitting the expected integrated variance of individual futures returns (Table 10 (Panel B)). The model does not exhibit a directional bias in generating deviations from actual values. While it underpredicts futures return volatilities of euro and yen, it overpredicts the remaining four. The absolute errors range between 99 bps to 196 bps.

Consistent with our assessment of model accuracy, the Mincer-Zarnowitz type regressions of futures return variances support an average (across the cross-section of six names) $\overline{R}^2$ of 65%. Furthermore, the model prices the futures on the dollar index with average absolute errors of \$0.17.

5.6. Summary of the findings from the estimated model and benchmarking

All in all, the model helps to improve our understanding of return properties of geometrically weighted currency composite, its futures and options, and volatilities.

First, the estimated model supports negative risk premiums of OTM call options. This means that the state-contingent risk premiums associated with a rise in the futures are negative, and that the dollar index

futures align with the properties of a safe haven security. Second, the estimated model generates plausible implications for the panel of option prices on futures on the dollar index, the $\text{VIX}_t^{\text{usd}|\text{index}}$, and the return volatilities of the dollar index futures and its related six individual futures contracts.

Third, our framework provides evidence for a source of negative risk premiums of dollar index futures return volatility, since investors dislike an increase in volatility of pricing kernels. Furthermore, the positive relationship between V_t and $\frac{F_t^{T_F,\text{usd}|\text{index}}}{S_t^{\text{usd}|\text{index}}}$ in equation (15) implies that disregarding stochastic volatility can lead to unrealistic empirical results, particularly when generating option prices for the dollar index futures underlier. Combining both theoretical and empirical results from our assessment exercises, we conclude that our findings support the quantitative implications of the estimated model.

6. Conclusion

The dollar index is a geometrically weighted currency composite — a joint claim on six currencies — with the U.S. dollar as the reference and settles in U.S. dollars. In this construction, a rise in the dollar index is associated with a dollar appreciation. This paper investigates, theoretically and empirically, the behavior of futures on the dollar index and options on futures on the dollar index.

This paper poses new questions concerning the modeling of this market. For example, what are the attributes of state-contingent risk premiums for dollar index depreciations and appreciations? Are call option risk premiums more negative at higher strikes? Additionally, what could be the forces that influence option risk premiums? Analyses of data from 08/03/2001 to 08/07/2020 are used to explore these questions.

The study of options on dollar index futures yields new results. Specifically, we determine that the average excess returns of out-of-the-money call options (closest to 20- and 10-delta) are reliably negative. In contrast, the average excess returns of out-of-the-money put options are statistically insignificant. Additionally, there is evidence to suggest that the average excess returns of call options are more negative farther out-of-the-money, as indicated by bootstrap-based evidence which is statistically significant.

This paper distinguishes itself from others by introducing a model of dollar index futures and options,

estimating the model by Kalman filter and quasi-maximum likelihood, testing the model against the data on options, and examining the agreement with empirical properties. This model results in tractable formulations for both futures and option prices, as well as an analytical representation for $\mathrm{VIX}_t^{\text{dollar index}}$.

Our analysis motivates estimated models of exchange rate behavior to be aligned with the data dimensions of futures and options on the dollar index.[1] The model is essential in reflecting the negative risk premiums of out-of-the-money calls on futures on the dollar index and producing reliably more negative risk premiums at higher strikes. Further, its empirical performance metrics show that the model can be useful in understanding the variations in the price of option portfolios related to volatility. Additionally, the model provides option prices that are consistent with the data counterparts.

[1]Currently, the majority of investigations focus on individual currencies. Examples include exercises by Carr and Wu (2007), Burnside, Eichenbaum, Kleshchelski, and Rebelo (2011), Jurek (2014), Lustig, Roussanov, and Verdelhan (2014), Farhi, Fraiberger, Gabaix, Ranciere, and Verdelhan (2015), Jurek and Xu (2016), Colacito, Croce, Gavazzoni, and Ready (2018), Dahlquist, Heyerdahl-Larsen, Pavlova, and Penasse (2022), and Greenwood, Hanson, Stein, and Sunderam (2023). However, our study differs in that we specifically examine the risk premiums of options (and option pricing) on futures for a currency index, the dollar index. This distinction, in the context of an estimated currency model, sets our study apart from others in the field, as noted in Table 5.

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Table 1

Excess returns of options on futures on the dollar index

The dollar index is expressed as the currency composite value per unit of U.S. dollar. Let $F_t^{T_F,\text{usd}|\text{index}}$ be the price of the futures contract that underlies the dollar index construction. Reported are the return properties of options written on futures on the dollar index over option expiration cycles. The sample period is from 08/03/2001 to 08/07/2020, with 229 monthly option expiration cycles. With the U.S. dollar as the reference, we compute the excess returns of OTM puts and calls, as follows:

$$\underbrace{\mathbb{r}^{\text{put}}_{\{t\to T_O\}}[k]}_{\text{monthly, \%}} = \frac{\max(K - F_{T_O}^{T_F,\text{usd}|\text{index}}, 0)}{\text{put}_t[K]} - R_t^{\text{us}}, \text{ where } K \text{ corresponds to } k = \log(\frac{K}{F_t^{T_F,\text{usd}|\text{index}}}) = -1\%, -2\%, \text{ and} -3\%,$$

$$\underbrace{\mathbb{r}^{\text{call}}_{\{t\to T_O\}}[k]}_{\text{monthly, \%}} = \underbrace{\frac{\max(F_{T_O}^{T_F,\text{usd}|\text{index}} - K, 0)}{\text{call}_t[K]}}_{\text{in U.S. dollars}} - R_t^{\text{us}}, \text{ where } K \text{ corresponds to } k = \log(\frac{K}{F_t^{T_F,\text{usd}|\text{index}}}) = 1\%, 2\%, \text{ and } 3\%,$$

where K is the strike price. For each date t, R_t^{us} is the gross U.S. LIBOR rate. The source of the matched futures and option prices is Barchart. We report the sample mean and the 95% bootstrap confidence intervals. Bootstrap confidence intervals on mean option excess returns that do *not* bracket zero are indicated by "•." We employ a stationary bootstrap procedure with 10,000 draws. The reported Sharpe ratios are the average option excess returns divided by their standard deviations, which we scale by $\sqrt{\frac{365}{30}}$ to annualize. $\mathbb{1}_{\{\mathbb{r}>0\}}$ represents the number (out of 229) of option excess returns that are positive. σ^{Black} is the option-implied volatility extracted using the model of Black (1976). The put (call) delta is $-e^{-r(T_O-t)}\mathcal{N}(-d_1)$ ($e^{-r(T_O-t)}\mathcal{N}(d_1)$), where $d_1 = \frac{1}{\sigma\sqrt{(T_O-t)}}\{-\log(k) + \frac{1}{2}\sigma^2(T_O-t)\}$ (r is the annualized LIBOR rate input and $T_O - t = \frac{30}{365}$). Shown also are properties of the excess returns of futures on the dollar index, computed as $\mathbb{r}^{\text{futures,usd}|\text{index}}_{\{t\to T_O\}} = \frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}} - 1$, together with those of each of the currency constituents (over option expiration cycles (annualized, and in %)). The p-values, denoted by NW[p], rely on the Newey-West HAC estimator (with the lag selected automatically). SD is standard deviation (annualized,%)

Panel A: Excess returns of options on futures on the dollar index (over 30 days, %)

	Delta (%)	Moneyness	Mean (monthly)	95% Bootstrap CI ⌊Lower	Upper⌋	(NW[p])	$\mathbb{1}_{\{\mathbb{r}>0\}}$ (out of 229)	Sharpe Ratio (annualized)	σ^{Black} (%)
Puts on dollar index futures	-10	3% OTM	-13	⌊−57	26⌋	(0.488)	22	-0.13	8.6
Puts on dollar index futures	-20	2% OTM	22	⌊−44	71⌋	(0.420)	40	0.17	8.3
Puts on dollar index futures	-35	1% OTM	9	⌊−24	38⌋	(0.436)	63	0.14	8.2
Straddle on dollar index futures			-4	⌊−13	5⌋	(0.464)	91	-0.2	8.1
Calls on dollar index futures	35	1% OTM	**-21**	⌊−45	1⌋	(0.037)	51	-0.41	8.3
Calls on dollar index futures	20	2% OTM	**-30•**	⌊−62	−3⌋	(0.030)	28	-0.46	8.6
Calls on dollar index futures	10	3% OTM	**-53•**	⌊−84	−28⌋	(0.000)	14	-0.83	9.0

Panel B: Excess returns of currency futures (annualized, %)

	Dollar is Reference	Mean (annualized)	95% Bootstrap CI ⌊Lower	Upper⌋	(NW[p])	SD	$\mathbb{1}_{\{\mathbb{r}>0\}}$ (out of 229)	Sharpe Ratio (annualized)
Futures	$\mathbb{r}^{\text{futures,usd}\|\text{index}}_{\{t\to T_O\}}$	-1.42	⌊−4.35	1.53⌋	(0.458)	7.9	112	-0.18
Futures	$\mathbb{r}^{\text{futures,usd}\|\text{eur}}_{\{t\to T_O\}}$	-0.53	⌊−4.01	2.90⌋	(0.818)	9.2	119	-0.06
Futures	$\mathbb{r}^{\text{futures,usd}\|\text{jpy}}_{\{t\to T_O\}}$	1.25	⌊−2.49	4.94⌋	(0.591)	9.7	109	0.14
Futures	$\mathbb{r}^{\text{futures,usd}\|\text{gbp}}_{\{t\to T_O\}}$	0.40	⌊−3.33	3.95⌋	(0.859)	9.5	105	0.04
Futures	$\mathbb{r}^{\text{futures,usd}\|\text{cad}}_{\{t\to T_O\}}$	-0.52	⌊−3.88	2.78⌋	(0.795)	8.8	111	-0.07
Futures	$\mathbb{r}^{\text{futures,usd}\|\text{sek}}_{\{t\to T_O\}}$	0.11	⌊−4.33	4.51⌋	(0.967)	11.7	118	0.01
Futures	$\mathbb{r}^{\text{futures,usd}\|\text{chk}}_{\{t\to T_O\}}$	-1.21	⌊−5.13	2.64⌋	(0.560)	10.2	110	-0.12

Table 2

Gauging collective concerns about U.S. dollar depreciations and appreciations

Reported are the option risk premium differentials at different moneyness. OTM puts have moneyness $k = \log(\frac{K}{F_t^{T_F,\text{usd|index}}}) < 0$, and OTM calls have moneyness $k = \log(\frac{K}{F_t^{T_F,\text{usd|index}}}) > 0$. We examine the following hypotheses:

(i) The risk premium of a call option of moneyness k is higher than the risk premium of a call option of lower moneyness.

(ii) The risk premium of a call option of moneyness $k > 0$ is higher than the risk premium of a put option of the same moneyness $k < 0$.

We jointly bootstrap — via stationary or circular block bootstrap procedures — the returns of the options with replacement and report the empirical frequency underlying each hypothesis. A low p-value implies rejection of the hypothesis. We perform 10,000 bootstraps. The sample is from 08/03/2001 to 08/07/2020, with 229 option expiration cycles.

	***p*-values from bootstrap procedures**	
	Stationary	Circular block
Panel A: Risk premium differentials across OTM call options		
Excess returns of a 2% OTM *call* minus that of a 1% OTM *call*	(0.042)	(0.053)
Excess returns of a 3% OTM *call* minus that of a 2% OTM *call*	(0.005)	(0.006)
Excess returns of a 3% OTM *call* minus that of a 1% OTM *call*	(0.001)	(0.001)
Panel B: Risk premium differentials across OTM call options and OTM put options		
Excess returns of a 3% OTM *call* minus that of a 3% OTM *put*	(0.068)	(0.067)
Excess returns of a 2% OTM *call* minus that of a 2% OTM *put*	(0.059)	(0.051)
Excess returns of a 1% OTM *call* minus that of a 1% OTM *put*	(0.097)	(0.084)

Table 3

Excess returns of options on futures on the dollar index from the standpoint of non-U.S. investors

All returns are denominated in the currency of the investor initiating the trade. Reported are the excess return properties — in local currency — of options written on futures on the dollar index over option expiration cycles. The holding period excess returns of OTM calls in local currency are constructed as follows:

$$\underbrace{\mathbb{r}^{\text{call, currency i}}_{\{t\to T_O\}}[k]}_{\substack{\text{based in currency i}\\ \text{(monthly, \%)}}} \equiv \underbrace{\frac{S^{\text{usd}|\text{currency i}}_{T_O}}{S^{\text{usd}|\text{currency i}}_{t}}}_{\text{currency i return}} \times \frac{\max(F^{T_F,\text{usd}|\text{index}}_{T_O} - K, 0)}{\text{call}_t[K]} - R^{\text{currency i}}_t, \text{ where } K \text{ corresponds to } k = 1\%,\ 2\%, \text{ and } 3\%.$$

In this calculation, K is the strike price, $F^{T_F,\text{usd}|\text{index}}_{T_O}$ is the price of the futures contract on the dollar index, and $R^{\text{currency i}}_t$ is the local currency LIBOR equivalent. We report the sample mean and the 95% bootstrap confidence intervals (employing a stationary bootstrap procedure with 1,000 draws).

		95% Bootstrap CI		Percentiles			Differences in mean call returns across OTM moneyness Bootstrap p-values		
							2% minus 1%	3% minus 2%	3% minus 1%
	Mean (monthly, %)	⌊Lower	Upper⌋	50th	75th	95th			
Panel A: Excess returns of calls on the dollar index initiated in the Eurozone euro									
Calls, 1% OTM	**-18**	⌊−43	4⌋	-100	-40	391	(0.049)	(0.005)	(0.000)
Calls, 2% OTM	**-27**	⌊−60	0⌋	-100	-100	399			
Calls, 3% OTM	**-51**	⌊−84	−25⌋	-100	-100	209			
Panel B: Excess returns of calls on the dollar index initiated in the Japanese yen									
Calls, 1% OTM	**-19**	⌊−44	2⌋	-100	-41	389	(0.048)	(0.005)	(0.000)
Calls, 2% OTM	**-29**	⌊−61	−1⌋	-100	-100	359			
Calls, 3% OTM	**-52**	⌊−85	−26⌋	-100	-100	186			
Panel C: Excess returns of calls on the dollar index initiated in the British pound									
Calls, 1% OTM	**-18**	⌊−43	4⌋	-100	-42	393	(0.047)	(0.005)	(0.005)
Calls, 2% OTM	**-28**	⌊−60	0⌋	-100	-100	401			
Calls, 3% OTM	**-51**	⌊−84	−24⌋	-100	-100	199			
Panel D: Excess returns of calls on the dollar index initiated in the Canadian dollar									
Calls, 1% OTM	**-19**	⌊−43	3⌋	-100	-42	388	(0.045)	(0.005)	(0.000)
Calls, 2% OTM	**-29**	⌊−61	−1⌋	-100	-100	392			
Calls, 3% OTM	**-52**	⌊−85	−26⌋	-100	-100	200			
Panel E: Excess returns of calls on the dollar index initiated in the Swedish krona									
Calls, 1% OTM	**-18**	⌊−43	5⌋	-100	-41	398	(0.048)	(0.005)	(0.001)
Calls, 2% OTM	**-27**	⌊−60	1⌋	-100	-100	405			
Calls, 3% OTM	**-51**	⌊−85	−24⌋	-100	-100	207			
Panel F: Excess returns of calls on the dollar index initiated in the Swiss franc									
Calls, 1% OTM	**-18**	⌊−43	4⌋	-100	-42	390	(0.049)	(0.004)	(0.000)
Calls, 2% OTM	**-28**	⌊−60	0⌋	-100	-100	392			
Calls, 3% OTM	**-51**	⌊−84	−24⌋	-100	-100	206			

Table 4

Partitioned average excess returns of call options and put options over expiration cycles

Reported are the properties of partitioned average excess returns of calls and puts on futures on the dollar index, computed over option expiration cycles. The sample period is 08/03/2001 to 08/07/2020, with 229 monthly option expiration cycles. We compute the excess returns of OTM options as in the note to Table 1. The following is the regression specification for calls and puts:

$$\underbrace{\mathbb{r}^{\text{option}}_{\{t\to T_O\}}[k]}_{\text{monthly, \%}} = \eta^{\text{option}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{pessimistic}}\}}\times\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{pessimistic}}\}}+\eta^{\text{option}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{normal}}\}}\times\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{normal}}\}}+\eta^{\text{option}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{optimistic}}\}}\times\mathbb{1}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{optimistic}}\}}+\underbrace{\varepsilon_{\{t\to T_O\}}}_{\text{error term}}.$$

We use proxies for the variable $\mathfrak{s}$, known at the beginning of the option expiration cycles, and the variable construction is described in the main text. We report the coefficients $\eta^{\text{option}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{pessimistic}}\}}$, $\eta^{\text{option}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{normal}}\}}$, and $\eta^{\text{option}}_{\{\mathcal{F}_t\in\mathfrak{s}_{\text{optimistic}}\}}$ (the mean in each state). The p-values, denoted by NW[p], rely on the Newey-West HAC estimator (with the lag selected automatically).

			Call options			**Put options**	
	Condition	States	**Mean** (monthly) (%)	(**NW[p]**)		**Mean** (monthly) (%)	(**NW[p]**)
Panel A: Currency volatility ($\text{Volatility}^{\text{currencies}}_t$)							
Calls, 1% OTM	Pessimistic	High $\text{Volatility}^{\text{currencies}}_t$	$\boxed{-25}$	(0.09)	Puts, 1% OTM	-14	(0.45)
	Normal	Medium $\text{Volatility}^{\text{currencies}}_t$	-20	(0.31)		11	(0.65)
	Optimistic	Low $\text{Volatility}^{\text{currencies}}_t$	-18	(0.57)		31	(0.42)
Calls, 2% OTM	Pessimistic	High $\text{Volatility}^{\text{currencies}}_t$	$\boxed{-42}$	(0.01)	Puts, 2% OTM	-9	(0.73)
	Normal	Medium $\text{Volatility}^{\text{currencies}}_t$	-30	(0.20)		9	(0.79)
	Optimistic	Low $\text{Volatility}^{\text{currencies}}_t$	-20	(0.64)		65	(0.41)
Calls, 3% OTM	Pessimistic	High $\text{Volatility}^{\text{currencies}}_t$	$\boxed{-59}$	(0.00)	Puts, 3% OTM	-17	(0.60)
	Normal	Medium $\text{Volatility}^{\text{currencies}}_t$	-64	(0.00)		-9	(0.81)
	Optimistic	Low $\text{Volatility}^{\text{currencies}}_t$	-37	(0.30)		-11	(0.80)
Panel B: Change in the price of volatility contract on futures on the dollar index ($\Delta\text{VOL}^{\text{dollar index}}_t$)							
Call, 1% OTM	Pessimistic	High $\Delta\text{VOL}^{\text{dollar index}}_t$	$\boxed{-30}$	(0.04)	Puts, 1% OTM	-9	(0.65)
	Normal	Medium $\Delta\text{VOL}^{\text{dollar index}}_t$	-18	(0.32)		-6	(0.79)
	Optimistic	Low $\Delta\text{VOL}^{\text{dollar index}}_t$	-14	(0.64)		43	(0.22)
Call, 2% OTM	Pessimistic	High $\Delta\text{VOL}^{\text{dollar index}}_t$	$\boxed{-40}$	(0.02)	Puts, 2% OTM	-12	(0.65)
	Normal	Medium $\Delta\text{VOL}^{\text{dollar index}}_t$	-34	(0.15)		3	(0.93)
	Optimistic	Low $\Delta\text{VOL}^{\text{dollar index}}_t$	-18	(0.67)		73	(0.34)
Call, 3% OTM	Pessimistic	High $\Delta\text{VOL}^{\text{dollar index}}_t$	$\boxed{-52}$	(0.01)	Puts, 3% OTM	-24	(0.43)
	Normal	Medium $\Delta\text{VOL}^{\text{dollar index}}_t$	-50	(0.11)		-11	(0.76)
	Optimistic	Low $\Delta\text{VOL}^{\text{dollar index}}_t$	-58	(0.02)		-3	(0.95)

Table 5

Distinct aspects of our paper in relation to 32 recent studies in the currency market

We adopt the following classification: Yes (No) means that the considered study satisfies (does not satisfy) the stated feature.

	Data on dollar index futures and options	Examine models of options on futures	Consider models of dollar index VIX	Key features of extant studies and their relation to our paper with estimated model and options on futures on the dollar index. Estimated model supports negative risk premiums of OTM call options. Our estimation approach is Kalman filtering. Estimated model produces plausible implications for the panel of dollar index option prices and the $\mathrm{VIX}_t^{\mathrm{usd}\mid\mathrm{index}}$.
This paper	Yes	Yes	Yes	Collective dollar concerns are imputed from dollar index options.
Carr and Wu (2007)	No	No	No	Focus on the issue of currency risk-reversals and its modeling.
Pavlova and Rigobon (2007)	No	No	No	Model of comovements: stock, bond, and currencies. No options.
Bakshi, Carr, and Wu (2008)	No	No	No	Model stochastic discount factors to produce currency skewness.
Verdelhan (2010)	No	No	No	Model with consumption and habits. No options.
Colacito and Croce (2011)	No	No	No	Long-run risks model for currencies. No options.
Burnside et. el. (2011)	No	No	No	Currency carry trade profits reflect a peso problem.
Lustig, Roussanov, and Verdelhan (2011)	No	No	No	Slope factor explains the cross-section of currency returns. No options.
Bansal and Shaliastovich (2013)	No	No	No	Framework for studying long-run risks. No options.
Colacito and Croce (2013)	No	No	No	Reconcile the low correlation between consumption and currencies.
Hassan (2013)	No	No	No	Size of economy explains currency returns cross-section. No options.
Maggiori (2013)	No	No	No	U.S. dollar earns a safety premium. No options.
Jurek (2014)	No	No	No	Examines returns of crash-hedged portfolios.
Lettau, Maggiori, and Weber (2014)	No	No	No	Model of downside risks. No options.
Lustig, Roussanov, and Verdelhan (2014)	No	No	No	Model of countercyclical risk premiums. No options.
Farhi et. el. (2015)	No	No	No	Disaster risk is priced in currency markets.
Della Corte, Sarno, and Ramadorai (2016)	No	No	No	Empirical study of currency volatility risk premiums.
Farhi and Gabaix (2016)	No	No	No	Develop a link between exchange rates and disasters.
Graveline and Joslin (2016)	No	No	No	Model joint dynamics of currencies and swap term structure. No options.
Jurek and Xu (2016)	No	No	No	Estimate currency risk premiums using options.
Mueller, Stathopoulos, and Vedolin (2017)	No	No	No	Empirical properties of currency return correlations imputed from options.
Ready, Roussanov, and Ward (2017)	No	No	No	Connect currency carry trade to commodity trade. No options.
Zviadadze (2017)	No	No	No	Currency carry returns is tied to macroeconomic risks. No options.
Chernov, Graveline, and Zviadadze (2018)	No	No	No	Discrete-time empirical model (four currencies) with jumps and volatility.
Colacito, Croce, Gavazzoni, and Ready (2018)	No	No	No	Model of exposures about global growth news shocks. No options.
Bakshi, Gao, and Panayotov (2021)	No	No	No	Cross-section of options on 45 spot currencies. Modeling of currency moments.
Jiang, Krishnamurthy, and Lustig (2021)	No	No	No	U.S. dollar's valuation is tied to convenience yields. No options.
Maurer and Tran (2021)	No	No	No	Model for currency volatility and currency risk premia. No options.
Zhang (2021)	No	No	No	Smooth exchange rates coincide with correlated equity returns. No options.
Aloosh and Bekaert (2022)	No	No	No	Factor model of currencies. No options.
Bakshi and Londono (2022)	No	No	No	Daily options panel of 15 spot currencies. Stochastic intensity model.
Dahlquist et. el. (2022)	No	No	No	Model with habits matches currency risk premiums. No options.
Greenwood et. el. (2023)	No	No	No	Supply of long-term bonds connects to currency fluctuations. No options.

Table 6

Estimation results based on the quasi-maximum likelihood and Kalman filter

This table provides the quasi-maximum likelihood estimates and standard errors. We employ a Kalman filter with seven measurement equations (as seen in equation (23)) and one transition equation for V_t (as seen in equation (24)). The data inputs include (i) futures and spot prices of the dollar index and (ii) realized variance of futures returns (the sum of daily squared returns over option expiration cycles) corresponding to the Eurozone euro, the Japanese yen, the British pound, the Canadian dollar, the Swedish krona, and the Swiss franc. The estimation is done over option expiration cycles. The reported $\mathfrak{L}$ is the log-likelihood, ACF_1 is the first-order autocorrelation and SD is standard deviation.

	Panel A: Parameters of V_t			**Panel B: Pricing kernel parameters**							
	κ_v	θ_v	σ_v	α_{us}	$\alpha_{eurozone}$	α_{japan}	α_{uk}	α_{canada}	α_{sweden}	$\alpha_{switzerland}$	$\mathfrak{L}$
Estimate	3.004	0.906	1.192	1.268	1.098	1.101	1.097	1.098	1.046	1.085	5172
Standard Error	0.967	0.259	0.128	0.0698	0.031	0.030	0.028	0.029	0.019	0.021	

	Panel C: Dispersion of measurement errors						
	σ^f	$\sigma^{rv}_{eurozone}$	σ^{rv}_{japan}	σ^{rv}_{uk}	σ^{rv}_{canada}	σ^{rv}_{sweden}	$\sigma^{rv}_{switzerland}$
Estimate	0.0205	0.0034	0.0064	0.0082	0.0055	0.0068	0.0247
Standard Error	0.0010	0.0002	0.0003	0.0004	0.0003	0.0005	0.0012

	Panel D: Properties of extracted $\{\sqrt{V_t} : t = 1, \ldots, T\}$									
	Mean	SD	Min.	5th	25th	50th	75th	95th	Max.	ACF_1
Extracted $\sqrt{V_t}$ (%) (annualized)	52.6	18.8	24.7	31.2	41.0	49.5	58.8	86.3	160.5	0.76

Table 7

Model fitting errors for options on futures on the dollar index

At the beginning of the option expiration cycle, the model price of options on futures on the dollar index is computed as follows:

$$\underbrace{\text{call}_t[K^{\text{usd}|\text{index}}]}_{\text{usd is reference}} = \frac{F_t^{T_F,\text{usd}|\text{index}}}{R_t^{\text{us}}}\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\mathbb{x}^{\text{index}|\text{usd}}_{\{t\to T_O\}}}) \times \{\frac{1}{2} - \frac{1}{\pi}\int_0^\infty \text{Re}[\frac{e^{-i\phi k^{\text{index}|\text{usd}}}\frac{\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{\{i\phi-1\}\mathbb{x}^{\text{index}|\text{usd}}})}{\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\mathbb{x}^{\text{index}|\text{usd}}})}}{i\phi}]\,d\phi\}$$

$$- \frac{K^{\text{usd}|\text{index}}}{R_t^{\text{us}}}\{\frac{1}{2} - \frac{1}{\pi}\int_0^\infty \text{Re}[\frac{e^{-i\phi k^{\text{index}|\text{usd}}}\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{i\phi\mathbb{x}^{\text{index}|\text{usd}}})}{i\phi}]\,d\phi\},$$

and

$$\underbrace{\text{put}_t[K^{\text{usd}|\text{index}}]}_{\text{usd is reference}} = \frac{K^{\text{usd}|\text{index}}}{R_t^{\text{us}}}\{\frac{1}{2} + \frac{1}{\pi}\int_0^\infty \text{Re}[\frac{e^{-i\phi k^{\text{index}|\text{usd}}}\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{i\phi\mathbb{x}^{\text{index}|\text{usd}}})}{i\phi}]d\phi\}$$

$$- \frac{F_t^{T_F,\text{usd}|\text{index}}}{R_t^{\text{us}}}\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\mathbb{x}^{\text{index}|\text{usd}}})\{\frac{1}{2} + \frac{1}{\pi}\int_0^\infty \text{Re}[\frac{e^{-i\phi k^{\text{index}|\text{usd}}}\frac{\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{\{i\phi-1\}\mathbb{x}^{\text{index}|\text{usd}}})}{\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\mathbb{x}^{\text{index}|\text{usd}}})}}{i\phi}]d\phi\}.$$

The expressions for $\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{\{i\phi-1\}\mathbb{x}^{\text{index}|\text{usd}}_{\{t\to T_O\}}})$, $\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{i\phi\mathbb{x}^{\text{index}|\text{usd}}_{\{t\to T_O\}}})$, and $\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\mathbb{x}^{\text{index}|\text{usd}}_{\{t\to T_O\}}})$ are obtained from equation (B12) of Appendix B. In the calculations, we use the parameter estimates and the extracted values of V_t (as in Table 6). Reported are the results from Mincer-Zarnowitz type regressions for option implied volatility:

$$\log(\{\sigma_t^{\text{Black}}\}^{\text{actual}}) = \beta_0 + \beta_1 \log(\{\sigma_t^{\text{Black}}\}^{\text{model}}) + e_t.$$

The regression intercept is not reported and significant slope coefficients are marked with "•." CORR refers to the correlation between the actual and model-generated time-series. Reported are the following model error measures:

$$\underbrace{\text{e}_t}_{\text{model errors}} = \text{Model}_t - \text{Actual}_t, \qquad \underbrace{\frac{1}{T}\sum_{t=1}^{T}\text{e}_t}_{\text{average errors}} \qquad \text{and} \qquad \underbrace{\frac{1}{T}\sum_{t=1}^{T}|\text{e}_t|}_{\text{average absolute errors}}.$$

σ^{Black} is the option-implied volatility extracted using the model of Black (1976).

		Errors in option prices		**Errors in σ^{Black}**		**Mincer-Zarnowitz regressions** ($\log(\sigma^{\text{Black}})$)		
		Average ($)	Absolute ($)	Average (%)	Absolute (%)	Slope (β_1)	$\overline{R}^2$ (%)	CORR (actual, model)
Puts on dollar index futures	3% OTM	-0.044	0.077	-1.79	2.30	0.42•	59	0.81
Puts on dollar index futures	2% OTM	-0.032	0.096	-0.68	1.49	0.56•	66	0.83
Puts on dollar index futures	1% OTM	0.018	0.106	0.18	1.15	0.76•	68	0.84
Puts on dollar index futures	0% OTM	0.071	0.119	0.70	1.18	0.86•	69	0.84
Calls on dollar index futures	0% OTM	0.142	0.162	1.41	1.61	0.79•	69	0.83
Calls on dollar index futures	1% OTM	0.158	0.169	1.67	1.79	0.84•	69	0.83
Calls on dollar index futures	2% OTM	0.141	0.151	1.80	1.93	0.85•	66	0.82
Calls on dollar index futures	3% OTM	0.108	0.121	1.82	2.04	0.84•	60	0.79

Table 8

Model-generated risk premiums of OTM call options and risk premiums of return volatility

At the beginning of the option expiration cycle, the risk premium of call on the dollar index futures is computed as

$$\text{Call risk premium}_{\{t\to T_O\}}[K^{\text{usd|index}}] \;=\; \underbrace{\frac{\mathbb{E}_t^{\mathbb{P}}(\max(F_{T_O}^{T_F,\text{usd|index}} - K^{\text{usd|index}},0))}{\text{call}_t[K^{\text{usd|index}}]}}_{\text{from model}} \;-\; R_t^{\text{us}}.$$

The expression for $\mathbb{E}_t^{\mathbb{P}}(\max(F_{T_O}^{T_F,\text{usd|index}} - K^{\text{usd|index}},0))$ is shown in (B16), whereas the call price is in (27). We use the parameter estimates and the extracted values of V_t (as in Table 6). Additionally, we compute the model-determined volatility risk premium in line with equation (30). Correspondingly, we compute the data-based average volatility risk premium as the time-series average of the log of the realized volatility divided by the options-inferred counterpart. Shown also are the lower and upper 95% bootstrap confidence intervals on the average volatility risk premium. SD is standard deviation.

	Properties of call option risk premiums (monthly, %)								
				Percentiles					
	Mean	SD	Min.	5th	25th	50th	75th	95th	Max.
Call risk premium$_{\{t\to T_O\}}[K^{\text{usd|index}}]$, 1% OTM	-32$^{\bullet}$	6	-62	-42	-34	-31	-27	-24	-20
Call risk premium$_{\{t\to T_O\}}[K^{\text{usd|index}}]$, 2% OTM	-36$^{\bullet}$	5	-65	-45	-38	-35	-32	-29	-26
Call risk premium$_{\{t\to T_O\}}[K^{\text{usd|index}}]$, 3% OTM	-40$^{\bullet}$	5	-67	-49	-42	-39	-37	-34	-31
Call risk premium: 3% OTM <u>minus</u> 1% OTM	-8.4	2.0	-12.7	-11.3	-9.8	-8.8	-7.4	-4.7	-3.3
(NW[p])	(0.000)								
	Properties of volatility risk premiums (annualized, %)								
Volatility risk premium$_{\{t\to T_O\}}$ (model)	-9.4$^{\bullet}$	1.2	-15.4	-11.3	-10.5	-8.8	-8.4	-8.1	-7.4
Volatility risk premium$_{\{t\to T_O\}}$ (data)									
Average	-10.6								
Bootstrap 95% confidence intervals	$\lfloor -13.9,\ -7.2 \rfloor$								

Table 9

Actual versus model-generated $\mathrm{VIX}_t^{\mathbf{usd|index}}$

All reported results rely on quasi-maximum likelihood estimates. We use Kalman filtering with seven measurement equations (as seen in equation (23)) and one transition equation for V_t (as seen in equation (24)). The data inputs are (i) futures and spot prices of the dollar index, and (ii) realized variance of futures returns (the sum of daily squared returns over option expiration cycles) corresponding to the Eurozone euro, the Japanese yen, the British pound, the Canadian dollar, the Swedish krona, and the Swiss franc. The estimation is done over option expiration cycles. From options on futures on the dollar index, we derive the actual $\{\mathrm{VIX}_t^{\mathrm{usd|index}}\}^2$ at the beginning of each option expiration cycle using equation (32). The model-implied $\{\mathrm{VIX}_t^{\mathrm{usd|index}}\}^2$ is based on equation (31). ACF_1 is the first-order autocorrelation and SD is standard deviation.

Panel A: Properties of actual and model-implied $\mathrm{VIX}_t^{\mathrm{usd|index}}$ **(annualized, %)**

	Mean	SD	Min.	Percentiles 5th	25th	50th	75th	95th	Max.	ACF_1
Actual $\mathrm{VIX}_t^{\mathrm{usd|index}}$	8.72	2.44	4.35	5.49	7.42	8.47	9.38	12.84	21.45	0.81
Model-implied $\mathrm{VIX}_t^{\mathrm{usd|index}}$	9.26	2.93	5.17	6.04	7.45	8.74	10.18	14.54	26.56	0.79

Panel B: Correlations with actual currency futures returns $(\mathbb{r}^{\mathrm{futures,usd|i}}_{\{t\to T_O\}})$

Currency i	Dollar Index	EUR	JPY	GBP	CAD	SEK	CHF
Log changes in:							
Actual $\mathrm{VIX}_t^{\mathrm{usd|index}}$	0.20	0.22	-0.02	0.21	0.13	0.17	0.20
Model-implied $\mathrm{VIX}_t^{\mathrm{usd|index}}$	0.21	0.19	-0.06	0.33	0.23	0.27	0.04

Table 10

Quantitative model assessments from return volatilities

Reported are the following model error measures:

$$\underbrace{\mathrm{e}_t}_{\text{model errors}} = \mathrm{Model}_t - \mathrm{Actual}_t, \qquad \underbrace{\frac{1}{T}\sum_{t=1}^{T}\mathrm{e}_t}_{\text{average errors}} \qquad \text{and} \qquad \underbrace{\frac{1}{T}\sum_{t=1}^{T}|\mathrm{e}_t|}_{\text{average absolute errors}}.$$

We report the model errors for (i) $\mathrm{VIX}_t^{\mathrm{usd}|\mathrm{index}}$ (annualized, %), (ii) dollar index futures return volatility ($\sqrt{\mathrm{rv}_t}$, annualized, %), (iii) individual futures return volatility ($\sqrt{\mathrm{rv}_t}$, annualized, %), and (iv) the futures on the dollar index (in U.S. dollars). The following Mincer-Zarnowitz type regressions are performed in log of the variable:

$$\log(\mathrm{Actual}_t) = \beta_0 + \beta_1 \log(\mathrm{Model}_t) + e_t.$$

The regression intercept is not reported and significant slope coefficients are marked with "•." CORR refers to the correlation between the actual and model-generated values of the variable (in logs).

	Model errors		**Mincer-Zarnowitz regressions (in logs)**		
	Average	Absolute	Slope (β_1)	$\overline{R}^2$ (%)	CORR (actual, model)
Panel A: Model performance (variables *not* used in the measurement equations)					
$\mathrm{VIX}_t^{\mathrm{usd}\|\mathrm{index}}$	0.54	1.41	0.68•	53	0.73
Dollar index futures return volatility	1.43	1.48	1.17•	86	0.92
Panel B: Model performance (variables used in the measurement equations)					
Eurozone euro futures return volatility	-0.11	0.99	1.21•	85	0.92
Japanese yen futures return volatility	-0.33	1.96	0.87•	42	0.65
British pound futures return volatility	0.28	1.57	0.95•	56	0.75
Canadian dollar futures return volatility	0.67	1.61	1.08•	61	0.78
Swedish krona futures return volatility	0.85	1.47	1.18•	85	0.92
Swiss franc futures return volatility	0.10	1.82	1.09•	61	0.78
Dollar index futures	-\$0.12	\$0.17	0.99•	100	1.0

Appendix

A. Appendix A: Proof of expression (15) for futures on the dollar index

For our characterizations, note that V_t — that affects pricing kernel volatilities — is governed by the following dynamics:

$$\begin{aligned}
\underbrace{d\mathrm{V}_t}_{\text{under }\mathbb{P}} &= (\theta_{\mathrm{v}} - \kappa_{\mathrm{v}}\,\mathrm{V}_t)\,dt \;+\; \sigma_{\mathrm{v}}\sqrt{\mathrm{V}_t}\,d\omega_t^{\mathbb{P}}, \\
\underbrace{d\mathrm{V}_t}_{\text{under }\mathbb{Q}^{\mathrm{us}}} &= (\theta_{\mathrm{v}} - \underbrace{\{\kappa_{\mathrm{v}} - \alpha_{\mathrm{us}}\,\sigma_{\mathrm{v}}\}}_{\text{adjustment under }\omega_t^{\mathbb{Q}^{\mathrm{us}}}}\,\mathrm{V}_t)\,dt \;+\; \sigma_{\mathrm{v}}\sqrt{\mathrm{V}_t}\,d\omega_t^{\mathbb{Q}^{\mathrm{us}}}, \;\; \text{where } \alpha_{\mathrm{us}}, \theta_{\mathrm{v}}, \kappa_{\mathrm{v}}, \text{ and } \sigma_{\mathrm{v}} \text{ are constants.} \quad \text{(A1)}
\end{aligned}$$

Ito's lemma applied to $\log(S_t^{\mathrm{index}|\mathrm{usd}})$ — using equations (11) and (12) — implies

$$\begin{aligned}
\underbrace{d\log(S_t^{\mathrm{index}|\mathrm{usd}})}_{\text{usd is nonreference}} &= (r_t^{\mathrm{us}} - r_t^{\mathrm{index}})\,dt \;+\; \frac{1}{2}\sum_{\mathrm{i}}(\mathbb{w}_{\mathrm{i}})(\mathbb{w}_{\mathrm{i}} - 1)\,\{\alpha_{\mathrm{i}} - \alpha_{\mathrm{us}}\}^2\,\mathrm{V}_t dt \\
&+ \frac{1}{2}\sum_{i}\sum_{j\neq i}(2\mathbb{w}_{\mathrm{i}}\,\mathbb{w}_{\mathrm{j}}) \times \{\alpha_{\mathrm{i}} - \alpha_{\mathrm{us}}\} \times \{\alpha_{\mathrm{j}} - \alpha_{\mathrm{us}}\}\mathrm{V}_t\,dt \\
&- \frac{1}{2}\Big\{\sum_{\mathrm{i}}\mathbb{w}_{\mathrm{i}}\,\{\alpha_{\mathrm{i}} - \alpha_{\mathrm{us}}\}\Big\}^2\,\mathrm{V}_t\,dt \\
&+ \sum_{\mathrm{i}}\mathbb{w}_{\mathrm{i}}\,\{\alpha_{\mathrm{i}} - \alpha_{\mathrm{us}}\}\sqrt{\mathrm{V}_t}\,d\omega_t^{\mathbb{Q}^{\mathrm{us}}}. \qquad \text{(A2)}
\end{aligned}$$

By an integration step,

$$\underbrace{\frac{S_{T_F}^{\mathrm{index}|\mathrm{usd}}}{S_t^{\mathrm{index}|\mathrm{usd}}}}_{\text{usd is nonreference}} = \exp((r_t^{\mathrm{us}} - r_t^{\mathrm{index}})\,(T_F - t)) \times \exp\Big(A\int_t^{T_F}\mathrm{V}_u\,du \;+\; B\int_t^{T_F}\sqrt{\mathrm{V}_u}\,d\omega_u^{\mathbb{Q}^{\mathrm{us}}}\Big), \qquad \text{(A3)}$$

for constants A and B, as displayed in equations (18) and (19). The futures price can be determined as

$$\begin{aligned}
\frac{\overbrace{S_t^{\mathrm{usd}|\mathrm{index}}}^{\text{usd is reference}}}{F_t^{T_F,\mathrm{usd}|\mathrm{index}}} &= \frac{\overbrace{F_t^{T_F,\mathrm{index}|\mathrm{usd}}}^{\text{usd is nonreference}}}{S_t^{\mathrm{index}|\mathrm{usd}}} = \frac{1}{S_t^{\mathrm{index}|\mathrm{usd}}}\mathbb{E}_t^{\mathbb{Q}^{\mathrm{us}}}(S_{T_F}^{\mathrm{index}|\mathrm{usd}}), \qquad \text{(A4)} \\
&= \exp((r_t^{\mathrm{us}} - r_t^{\mathrm{index}})\,(T_F - t)) \times \mathbb{E}_t^{\mathbb{Q}^{\mathrm{us}}}\Big(\exp(A_*\int_t^{T_F}\mathrm{V}_u\,du \;+\; B_*\int_t^{T_F}\sqrt{\mathrm{V}_u}\,d\omega_u^{\mathbb{Q}^{\mathrm{us}}})\Big). \qquad \text{(A5)}
\end{aligned}$$

Solving this expectation (using such sources as Friedman (1975)) subject to $F_{T_F}^{T_F,\mathrm{index}|\mathrm{usd}} = S_{T_F}^{\mathrm{index}|\mathrm{usd}}$ (and constants A_* and B_*) leads to the solution for the futures price on the dollar index shown in (15). ■

B. Appendix B: Expressions for option price and option risk premiums

For brevity of presentation, consider the following rearrangements:

$$\underbrace{k^{\text{usd}|\text{index}}}_{>0} = \log(\frac{K^{\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}) = \underbrace{\log(\frac{F_t^{T_F,\text{index}|\text{usd}}}{K^{\text{index}|\text{usd}}})}_{\text{reciprocal}} = -\log(\frac{K^{\text{index}|\text{usd}}}{F_t^{T_F,\text{index}|\text{usd}}}) \equiv \; -\underbrace{k^{\text{index}|\text{usd}}}_{<0}. \tag{B1}$$

Next define the log futures return with the U.S. dollar as the nonreference currency, as follows:

$$\underbrace{\mathbb{x}^{\text{index}|\text{usd}}_{\{t\to T_O\}}}_{\text{usd is nonreference}} \;\equiv\; \underbrace{\log(\frac{F_{T_O}^{T_F,\text{index}|\text{usd}}}{F_t^{T_F,\text{index}|\text{usd}}})}_{\text{log futures return}}. \tag{B2}$$

I. Relating option risk premiums to risk premiums on powers of log futures returns

The payoff of the call on futures on the dollar index has the following approximate representation:

$$\begin{aligned}
\underbrace{\mathbf{c}_{T_O}^{\text{payoff}}[k^{\text{usd}|\text{index}}]}_{\text{call payoff of dollar index}} &= F_t^{T_F,\text{usd}|\text{index}} \max(\{\frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}} - 1\} \;-\; \{\frac{K^{\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}} - 1\}, 0) \quad \longleftarrow \text{ subtracting and adding one} \\
&\approx F_t^{T_F,\text{usd}|\text{index}} \max(\log(\frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}) \;-\; \log(\frac{K^{\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}), 0) \quad \longleftarrow \text{ log approximation for small } T_O \\
&= \frac{1}{F_t^{T_F,\text{index}|\text{usd}}} \max(\log(\frac{K^{\text{index}|\text{usd}}}{F_t^{T_F,\text{index}|\text{usd}}}) \;-\; \log(\frac{F_{T_O}^{T_F,\text{index}|\text{usd}}}{F_t^{T_F,\text{index}|\text{usd}}}), 0) \quad \longleftarrow \text{ reordering the reference} \qquad \text{(B3)} \\
&= \frac{1}{F_t^{T_F,\text{index}|\text{usd}}} \max(k^{\text{index}|\text{usd}} \;-\; \mathbb{x}^{\text{index}|\text{usd}}_{\{t\to T_O\}}, 0). \qquad \text{(B4)}
\end{aligned}$$

Then, for $i = \sqrt{-1}$, we have — after a change of variable, shifting property, and specialization — the following building block identity (Bateman and Erdelyi (1954, 3.2.3, page 118)):

$$\max(k^{\text{index}|\text{usd}} \;-\; \mathbb{x}^{\text{index}|\text{usd}}_{\{t\to T_O\}}, 0) \;=\; \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{1}{(1+i\phi)^2}\, e^{\{1+i\phi\}\{k^{\text{index}|\text{usd}} \;-\; \mathbb{x}^{\text{index}|\text{usd}}_{\{t\to T_O\}}\}}\, d\phi. \tag{B5}$$

Aided by a Taylor series representation, one also obtains

$$e^{-\{1+i\phi\}\mathbb{x}} \;=\; 1 + (-1-i\phi)\,\mathbb{x} \;-\; \frac{1}{2}(\phi - i)^2\mathbb{x}^2 \;+\; \frac{1}{6}(-1-i\phi)^3\,\mathbb{x}^3 + \frac{1}{24}(\phi-i)^4\,\mathbb{x}^4 \;+\; O\left[\mathbb{x}^5\right]. \tag{B6}$$

Additionally, we assume that

$$\mathbb{E}_t^{\mathbb{P}}(e^{\varphi\mathbb{x}}) \;<\; \infty \quad \text{and} \quad \mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{\varphi\mathbb{x}}) \;<\; \infty. \quad \text{for parameter } \varphi \in \Re. \tag{B7}$$

Assuming that one can interchange the order of taking expectations, and the infinite summations of the terms in the series expansion of $e^{-\{1+i\phi\}\mathbb{x}}$, we deduce, in light of identity in equation (B5), that

$$\underbrace{\mathbb{E}_t^{\mathbb{P}}(\max(\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}} - k^{\text{index}|\text{usd}}, 0))}_{\text{usd is nonreference}} = \frac{1}{2\pi}\int_{-\infty}^{\infty} \text{Re}[\frac{e^{\{1+i\phi\}k^{\text{index}|\text{usd}}}}{(1+i\phi)^2}\mathbb{E}_t^{\mathbb{P}}\Big(1+(-1-i\phi)\,\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}} - \frac{1}{2}(\phi-i)^2\,\{\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}}\}^2 + \frac{1}{6}(-1-i\phi)^3\,\{\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}}\}^3 + \dots\Big)]\,d\phi + \dots$$

By the same sequence of steps, it holds that

$$\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(\max(\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}} - k^{\text{index}|\text{usd}}, 0)) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \text{Re}[\frac{e^{\{1+i\phi\}k^{\text{index}|\text{usd}}}}{(1+i\phi)^2}\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}\Big(1+(-1-i\phi)\,\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}} - \frac{1}{2}(\phi-i)^2\,\{\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}}\}^2 + \frac{1}{6}(-1-i\phi)^3\,\{\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}}\}^3 + \dots\Big)]\,d\phi + \dots$$

We then obtain as follows:

$$\begin{aligned}\mathbb{E}_t^{\mathbb{P}}(\max(\frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}} - \frac{K^{\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}, 0)) - \mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(\max(\frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}} - \frac{K^{\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}, 0)) &\approx \mathbb{g}^{(1)}[k^{\text{index}|\text{usd}}] \times \text{rp}_{\{t\to T_O\}}^{(1)} \\ &+ \mathbb{g}^{(2)}[k^{\text{index}|\text{usd}}] \times \text{rp}_{\{t\to T_O\}}^{(2)} \quad \text{(B8)} \\ &+ \mathbb{g}^{(3)}[k^{\text{index}|\text{usd}}] \times \text{rp}_{\{t\to T_O\}}^{(3)} + \dots,\end{aligned}$$

where $\text{rp}_{\{t\to T_O\}}^{(n)} \equiv \mathbb{E}_t^{\mathbb{P}}(\{\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}}\}^n) - \mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(\{\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}}\}^n)$ are risk premiums on powers of log futures returns. The expressions for $\mathbb{g}^{(n)}[\bullet]$, for $n = 1,\dots,3$, are presented below:

$$\mathbb{g}^{(1)}[k^{\text{index}|\text{usd}}] \equiv \frac{1}{2\pi}\int_{-\infty}^{\infty} \text{Re}[\frac{e^{\{1+i\phi\}k^{\text{index}|\text{usd}}}}{(1+i\phi)^2} \times (-1-i\phi)]\,d\phi, \quad \text{(B9)}$$

$$\mathbb{g}^{(2)}[k^{\text{index}|\text{usd}}] \equiv \frac{1}{2\pi}\int_{-\infty}^{\infty} \text{Re}[\frac{e^{\{1+i\phi\}k^{\text{index}|\text{usd}}}}{(1+i\phi)^2} \times (-\frac{1}{2}) \times (\phi-i)^2]\,d\phi \quad \text{and} \quad \text{(B10)}$$

$$\mathbb{g}^{(3)}[k^{\text{index}|\text{usd}}] \equiv \frac{1}{2\pi}\int_{-\infty}^{\infty} \text{Re}[\frac{e^{\{1+i\phi\}k^{\text{index}|\text{usd}}}}{(1+i\phi)^2} \times (\frac{1}{6}) \times (-1-i\phi)^3]\,d\phi. \quad \text{(B11)}$$

In particular, $\mathbb{g}^{(n)}[\bullet]$ vary with moneyness but are otherwise time-invariant. ■

II. Fourier transforms for log futures returns under $\mathbb{P}$ and $\mathbb{Q}$

The call price is discounted risk-neutral expectation. In this regard, for $\mathbb{x}_{\{t\to T_O\}}^{\text{index}|\text{usd}} = \log(\frac{F_{T_O}^{T_F,\text{index}|\text{usd}}}{F_t^{T_F,\text{index}|\text{usd}}})$, the following solution for the Fourier transform emerges:

$$\underbrace{H_t^{\mathbb{Q}^{\text{us}}}[\phi]}_{\text{under } \mathbb{Q}^{\text{us}}} \equiv \underbrace{\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}\left(e^{i\phi\,\mathbb{x}^{\text{index|usd}}_{\{t\to T_O\}}}\right)}_{\text{Fourier transform}} = \exp\left(\mathfrak{f}[T_O-t;\phi]+\mathfrak{g}[T_O-t;\phi]\,V_t\right), \tag{B12}$$

where for $\sigma_F[\tau] = \mathbb{b}[\tau]\,\sigma_{\text{v}} + \sum_{\text{i}} \mathbb{w}_{\text{i}}(\alpha_{\text{i}} - \alpha_{\text{us}})$ in equation (20), we have,

$$\begin{aligned}
\mathfrak{g}[T_O-t;\phi] &= \frac{2(A_f+\frac{1}{2}B_f^2)(1-e^{-\mho_f(T_O-t)})}{2\mho_f-(\mho_f-(\kappa_{\text{v}}-\alpha_{\text{us}}\sigma_{\text{v}}-\sigma_{\text{v}}B_f))(1-e^{-\mho_f(T_O-t)})},\\
\mathfrak{f}[T_O-t;\phi] &= -\frac{\theta_{\text{v}}}{\sigma_{\text{v}}^2}\Big((\mho_f-(\kappa_{\text{v}}-\alpha_{\text{us}}\sigma_{\text{v}}-\sigma_{\text{v}}B_f))(T_O-t)+2\log(1-\frac{(\mho_f-(\kappa_{\text{v}}-\alpha_{\text{us}}\sigma_{\text{v}}-\sigma_{\text{v}}B_f))(1-e^{-\mho_f(T_O-t)})}{2\mho_f})\Big),\\
\text{where } \mho_f[\phi] &= \sqrt{(\kappa_{\text{v}}-\alpha_{\text{us}}\sigma_{\text{v}}-\sigma_{\text{v}}B_f)^2-2\sigma_{\text{v}}^2(A_f+\frac{1}{2}B_f^2)},\\
B_f[\phi] &\equiv i\phi\,\sigma_F[\tau], \qquad \text{and}\\
A_f[\phi] &= -\frac{i\phi}{2}\{\sigma_F[\tau]\}^2.
\end{aligned}$$

Additionally noting that $\frac{M_t^{\text{us}}}{M_{T_O}^{\text{us}}\,e^{r_t^{\text{us}}(T_O-t)}} = e^{-\frac{1}{2}\alpha_{\text{us}}^2\int_t^{T_O}\mathrm{V}_u\,du-\alpha_{\text{us}}\int_t^{T_O}\sqrt{\mathrm{V}_u}\,d\omega_u^{\mathbb{Q}^{\text{us}}}}$ under $\mathbb{Q}^{\text{us}}$, we obtain

$$\begin{aligned}
\underbrace{H_t^{\mathbb{P}}[\phi]}_{\text{under } \mathbb{P}} \equiv \mathbb{E}_t^{\mathbb{P}}(e^{i\phi\,\mathbb{x}_{\{t\to T_O\}}}) &= \mathbb{E}_t^{\mathbb{P}}\Big(\frac{M_{T_O}^{\text{us}}\,e^{r_t^{\text{us}}(T_O-t)}}{M_t^{\text{us}}}\,\frac{e^{i\phi\,\mathbb{x}^{\text{index|usd}}_{\{t\to T_O\}}}}{\frac{M_T^{\text{us}}\,e^{r_t^{\text{us}}(T_O-t)}}{M_t^{\text{us}}}}\Big) = \mathbb{E}_t^{\mathbb{Q}^{\text{us}}}\Big(\frac{1}{\frac{M_{T_O}^{\text{us}}\,e^{r_t^{\text{us}}(T_O-t)}}{M_t^{\text{us}}}}\times e^{i\phi\,\mathbb{x}^{\text{index|usd}}_{\{t\to T_O\}}}\Big)\\
&= \mathbb{E}_t^{\mathbb{Q}^{\text{us}}}\Big(e^{-\frac{1}{2}\alpha_{\text{us}}^2\int_t^{T_O}\mathrm{V}_u\,du-\alpha_{\text{us}}\int_t^{T_O}\sqrt{\mathrm{V}_u}\,d\omega_u^{\mathbb{Q}^{\text{us}}}}\times e^{-\frac{i}{2}\phi\{\sigma_F[\tau]\}^2\int_t^{T_O}\mathrm{V}_u\,du+i\phi\,\sigma_F[\tau]\int_t^{T_O}\sqrt{\mathrm{V}_u}\,d\omega_u^{\mathbb{Q}^{\text{us}}}}\Big)\\
&= \exp\left(\mathfrak{f}[T_O-t;\phi]+\mathfrak{g}[T_O-t;\phi]\,V_t\right),
\end{aligned} \tag{B13}$$

where

$$\begin{aligned}
\mathfrak{g}[T_O-t;\phi] &= \frac{2(A_f+\frac{1}{2}B_f^2)(1-e^{-\mho_f(T_O-t)})}{2\mho_f-(\mho_f-(\kappa_{\text{v}}-\alpha_{\text{us}}\sigma_{\text{v}}-\sigma_{\text{v}}B_f))(1-e^{-\mho_f(T_O-t)})},\\
\mathfrak{f}[T_O-t;\phi] &= -\frac{\theta_{\text{v}}}{\sigma_{\text{v}}^2}\Big((\mho_f-(\kappa_{\text{v}}-\alpha_{\text{us}}\sigma_{\text{v}}-\sigma_{\text{v}}B_f))(T_O-t)+2\log(1-\frac{(\mho_f-(\kappa_{\text{v}}-\alpha_{\text{us}}\sigma_{\text{v}}-\sigma_{\text{v}}B_f))(1-e^{-\mho_f(T_O-t)})}{2\mho_f})\Big),\\
\text{where } \mho_f[\phi] &= \sqrt{(\kappa_{\text{v}}-\alpha_{\text{us}}\sigma_{\text{v}}-\sigma_{\text{v}}B_f)^2-2\sigma_{\text{v}}^2(A_f+\frac{1}{2}B_f^2)},\\
B_f[\phi] &\equiv i\phi\,\sigma_F[\tau] - \alpha_{\text{us}}, \text{ and}\\
A_f[\phi] &= -\frac{i}{2}\phi\{\sigma_F[\tau]\}^2-\frac{1}{2}\alpha_{\text{us}}^2.
\end{aligned}$$

We deduce $\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\{1+i\phi\}\mathbb{x}^{\text{index|usd}}_{\{t\to T_O\}}}) = H_t^{\mathbb{Q}^{\text{us}}}[\phi]\Big|_{\phi=-\{\frac{1}{i}+\phi\}}$ and $\mathbb{E}_t^{\mathbb{P}}(e^{-\{1+i\phi\}\mathbb{x}^{\text{index|usd}}_{\{t\to T_O\}}}) = H_t^{\mathbb{P}}[\phi]\Big|_{\phi=-\{\frac{1}{i}+\phi\}}$.

Relying on Fourier inversion techniques, we obtain the call price, shown as equation (27), as follows:

$$\underbrace{\text{call}_t[K^{\text{usd}|\text{index}}]}_{\text{usd is reference}} = F_t^{T_F,\text{usd}|\text{index}}\Big\{\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(e^{-\mathbb{x}^{\text{index}|\text{usd}}_{\{t\to T_O\}}}) \times \Big(\frac{1}{2} - \frac{1}{\pi}\int_0^\infty \text{Re}[\frac{e^{-i\phi k^{\text{index}|\text{usd}}} \times H_t^{*\mathbb{Q}^{\text{us}}}[\phi]}{i\phi}]\,d\phi\Big)$$

$$- e^{-k^{\text{index}|\text{usd}}_{\{t\to T_O\}}} \times \Big(\frac{1}{2} - \frac{1}{\pi}\int_0^\infty \text{Re}[\frac{e^{-i\phi k^{\text{index}|\text{usd}}} \times H_t^{\mathbb{Q}^{\text{us}}}[\phi]}{i\phi}]\,d\phi\Big)\Big\}, \quad \text{(B14)}$$

$$\text{where } H_t^{*\mathbb{Q}^{\text{us}}}[\phi] = \frac{H_t^{\mathbb{Q}^{\text{us}}}[\phi]\Big|_{\phi=\phi+i}}{H_t^{\mathbb{Q}^{\text{us}}}[\phi]\Big|_{\phi=-\frac{1}{i}}}, \quad \text{for} \quad H_t^{\mathbb{Q}^{\text{us}}}[\phi] \text{ in equation (B12).} \quad \text{(B15)}$$

We obtain the put price in equation (28) analogously. ■

We further have

$$\mathbb{E}_t^{\mathbb{P}}(\max(F_{T_O}^{T_F,\text{usd}|\text{index}} - K^{\text{usd}|\text{index}},0)) = F_t^{T_F,\text{usd}|\text{index}}\mathbb{E}_t^{\mathbb{P}}(e^{-\mathbb{x}^{\text{index}|\text{usd}}})\{\frac{1}{2} - \frac{1}{\pi}\int_0^\infty \text{Re}[\frac{e^{-i\phi k^{\text{index}|\text{usd}}}\frac{\mathbb{E}_t^{\mathbb{P}}(e^{i\phi\mathbb{x}^{\text{index}|\text{usd}} - \mathbb{x}^{\text{index}|\text{usd}}})}{\mathbb{E}_t^{\mathbb{P}}(e^{-\mathbb{x}^{\text{index}|\text{usd}}})}}{i\phi}]d\phi\}$$

$$- K^{\text{usd}|\text{index}}\{\frac{1}{2} - \frac{1}{\pi}\int_0^\infty \text{Re}[\frac{e^{-i\phi k^{\text{index}|\text{usd}}}\mathbb{E}_t^{\mathbb{P}}(e^{i\phi\mathbb{x}^{\text{index}|\text{usd}}})}{i\phi}]d\phi\}. \quad \text{(B16)}$$

We can now obtain obtain call option risk premiums. ■

C. Appendix C: Dollar index volatility risk premium

We compute the volatility risk premium on the futures on the dollar index as:

$$\text{Volatility risk premium}_{\{t\to T_O\}} = \log\frac{\sqrt{\frac{1}{T_O-t}\text{var}_t^{\mathbb{P}}[\{\frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}\}]}}{\sqrt{\frac{1}{T_o-t}\text{var}_t^{\mathbb{Q}^{\text{us}}}[\{\frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}\}]}}. \quad \text{(C1)}$$

Note the following relations:

$$\text{var}_t^{\mathbb{Q}^{\text{us}}}[\{\frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}\}] = \mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(\{\frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}}\}^2) - \{\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(\frac{F_{T_O}^{T_F,\text{usd}|\text{index}}}{F_t^{T_F,\text{usd}|\text{index}}})\}^2, \quad \text{(C2)}$$

$$= \mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(\exp(-2\log\frac{F_{T_O}^{T_F,\text{index}|\text{usd}}}{F_t^{T_F,\text{index}|\text{usd}}})) - \{\mathbb{E}_t^{\mathbb{Q}^{\text{us}}}(\exp(-\log\frac{F_{T_O}^{T_F,\text{index}|\text{usd}}}{F_t^{T_F,\text{index}|\text{usd}}}))\}^2, \quad \text{(C3)}$$

$$= \underbrace{H_t^{\mathbb{Q}^{\text{us}}}[\phi]\Big|_{\phi=\frac{-2}{i}}}_{\text{from equation (B12)}} - \underbrace{\{H_t^{\mathbb{Q}^{\text{us}}}[\phi]\Big|_{\phi=\frac{-1}{i}}\}^2}_{\text{from equation (B12)}}. \quad \text{(C4)}$$

Equation (C3) merely rearranges the reference.

The step for the next calculation is as follows:

$$\begin{aligned}
\mathrm{var}_t^{\mathbb{P}}[\{\frac{F_{T_O}^{T_F,\mathrm{usd}|\mathrm{index}}}{F_t^{T_F,\mathrm{usd}|\mathrm{index}}}\}] &= \mathbb{E}_t^{\mathbb{P}}(\{\frac{F_{T_O}^{T_F,\mathrm{usd}|\mathrm{index}}}{F_t^{T_F,\mathrm{usd}|\mathrm{index}}}\}^2) - \{\mathbb{E}_t^{\mathbb{P}}(\{\frac{F_{T_O}^{T_F,\mathrm{usd}|\mathrm{index}}}{F_t^{T_F,\mathrm{usd}|\mathrm{index}}}\})\}^2, && \text{(C5)}\\
&= \mathbb{E}_t^{\mathbb{P}}(\exp(-2\log\{\frac{F_{T_O}^{T_F,\mathrm{index}|\mathrm{usd}}}{F_t^{T_F,\mathrm{index}|\mathrm{usd}}}\})) - \{\mathbb{E}_t^{\mathbb{P}}(\exp(-\log\{\frac{F_{T_O}^{T_F,\mathrm{index}|\mathrm{usd}}}{F_t^{T_F,\mathrm{index}|\mathrm{usd}}}\}))\}^2, && \text{(C6)}\\
&= \underbrace{H_t^{\mathbb{P}}[\phi]\Big|_{\phi=\frac{-2}{i}}}_{\text{from equation (B13)}} - \underbrace{\{H_t^{\mathbb{P}}[\phi]\Big|_{\phi=\frac{-1}{i}}\}^2}_{\text{from equation (B13)}}. && \text{(C7)}
\end{aligned}$$

We have the expressions for implementation. ■

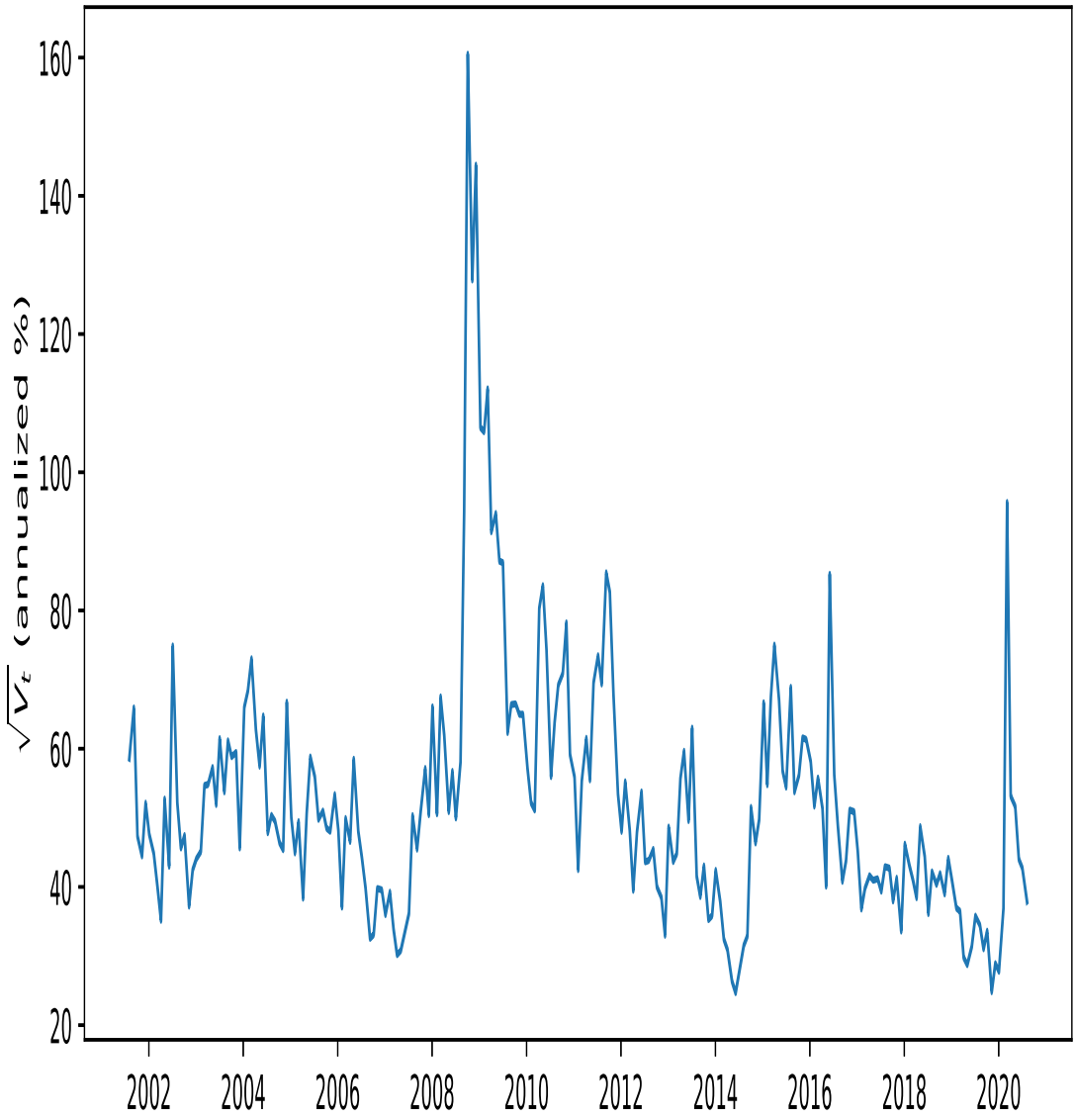

Figure 1. **Extracted values of $\sqrt{V_t}$ (common volatility component of pricing kernel growths)** We use Kalman filter and quasi-maximum likelihood for estimation. The estimation is done over option expiration cycles, and we use seven measurements (as shown in equation (23)). The properties of extracted $\{\sqrt{V_t} : t = 1, \ldots, T\}$ are displayed in Panel D of Table 6.

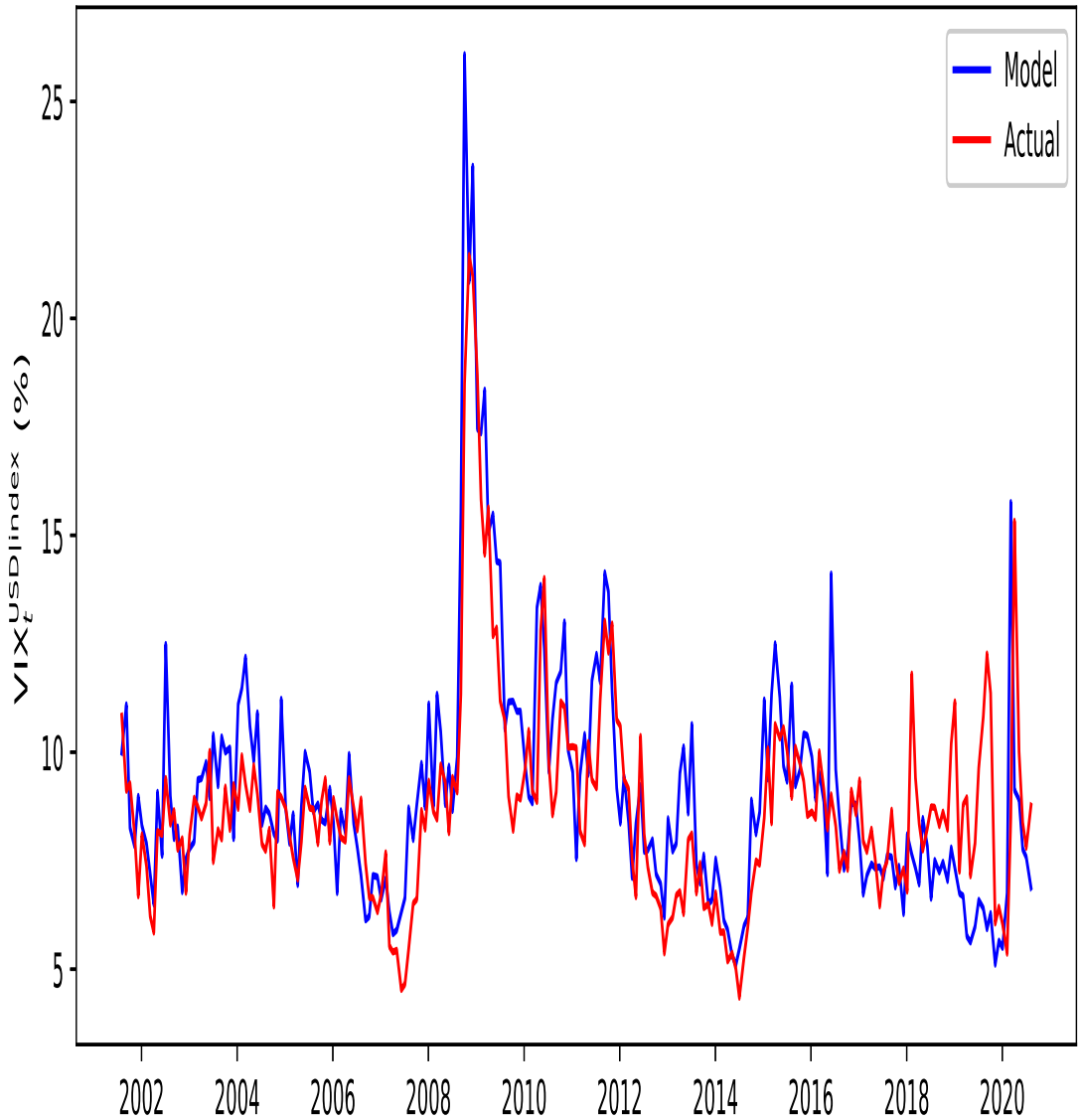

Figure 2. **Model and actual** $\mathrm{VIX}_t^{\mathrm{usd|index}}$

Parameters and extracted values of V_t are based on Kalman filter and quasi-maximum likelihood estimation. We use seven measurements (as shown in equation (23)). The model $\mathrm{VIX}_t^{\mathrm{usd|index}}$ is

$$\underbrace{\{\mathrm{VIX}_t^{\mathrm{usd|index}}\}^2}_{\text{model}} = \frac{\{\mathbb{b}[\tau]\,\sigma_{\mathrm{v}} + \sum_{\mathrm{i}} \mathbb{w}_{\mathrm{i}}(\alpha_{\mathrm{i}} - \alpha_{\mathrm{us}})\}^2}{T_O - t}\left(\frac{(T_O - t)\,\theta_{\mathrm{v}}}{\kappa_{\mathrm{v}} - \alpha_{\mathrm{us}}\,\sigma_{\mathrm{v}}} - \frac{(1 - e^{-\{\kappa_{\mathrm{v}} - \alpha_{\mathrm{us}}\,\sigma_{\mathrm{v}}\}\,(T_O - t)})}{\kappa_{\mathrm{v}} - \alpha_{\mathrm{us}}\,\sigma_{\mathrm{v}}}\{\frac{\theta_{\mathrm{v}}}{\kappa_{\mathrm{v}} - \alpha_{\mathrm{us}}\,\sigma_{\mathrm{v}}}\}\right)$$
$$+ \left(\frac{\{\mathbb{b}[\tau]\,\sigma_{\mathrm{v}} + \sum_{\mathrm{i}} \mathbb{w}_{\mathrm{i}}(\alpha_{\mathrm{i}} - \alpha_{\mathrm{us}})\}^2}{T_O - t} \times \frac{(1 - e^{-\{\kappa_{\mathrm{v}} - \alpha_{\mathrm{us}}\,\sigma_{\mathrm{v}}\}\,(T_O - t)})}{\kappa_{\mathrm{v}} - \alpha_{\mathrm{us}}\,\sigma_{\mathrm{v}}}\right)\mathrm{V}_t.$$

The actual $\mathrm{VIX}_t^{\mathrm{usd|index}}$ is computed as follows:

$$\underbrace{\{\mathrm{VIX}_t^{\mathrm{usd|index}}\}^2}_{\text{actual (from options)}} = \frac{2R_t^{\mathrm{us}}}{T_O - t}\left(\int_0^{F_t^{T_F,\mathrm{usd|index}}} \frac{\mathrm{put}_t[K]}{K^2}\,dK + \int_{F_t^{T_F,\mathrm{usd|index}}}^{\infty} \frac{\mathrm{call}_t[K]}{K^2}\,dK\right).$$